AI-ENABLED SMART LAMP POST FOR REALTIME EMERGENCY DETECTION, LOCALIZATION, AND MULTI-CHANNEL ALERTING

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

Safety systems deployed across densely populated urban and suburban regions predominantly operate in a response-oriented manner: an emergency usually comes to the attention of authorities only after a bystander observes it, locates a phone, and places a call. This project report presents SingaPen AI, a crisis-management system integrated directly into streetlight poles that combines physical sensing, on-device visual processing, and multiple independent wireless communication channels into a single economical unit. An MFRC522 RFID module paired with an ESP32 microcontroller allows an individual to trigger a linked identity alert, while a manual SOS button provides a parallel human-triggered pathway. Built-in redundancy is provided through a SIM800L GSM module and an SX1278 LoRa transceiver operating at 868 MHz, so that alert delivery does not depend on any single network. An onboard camera running Ultralytics YOLO11n performs edge-based weapon detection as an additional AI verification layer. All incoming data is routed to a Flask backend, with a Firebase Realtime Database collecting the live notification feed and a Leaflet-based dashboard mapping real-time incidents for dispatchers. The system accommodates two hardware forms a stationary Smart RFID Emergency Pole and a compact Portable Guardian SOS device both sharing the same alert schema and cloud backend. Campus deployment and testing at Rajalakshmi Institute of Technology, Chennai, measured sub-second end-to-end alert latency over WiFi (803 ms), reliable LoRa delivery over short range (100% packet success at 1 km, 85% at 5 km), and dependable weapon-class AI detection with a 5% false positive rate. These results indicate that integrating AI-powered emergency detection directly into existing streetlight infrastructure is an economical and effective approach to improving public safety response times.

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TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	**ii**
	LIST OF TABLES	**vi**
	LIST OF FIGURES	**vii**
1	**INTRODUCTION**	1
	1.1 BACKGROUND	
	1.2 PROBLEM STATEMENT	
	1.3 OBJECTIVES	
	1.4 SCOPE	
	1.5 SECURITY REQUIREMENTS / SECURITY CONSIDERATIONS	
	1.6 SDG ALIGNMENT	
	1.7 COMPARISON OF CONVENTIONAL AND PROPOSED RESPONSE	
	1.8 ORGANIZATION OF THE REPORT	
2	**LITERATURE REVIEW**	7
	2.1 EXISTING SYSTEM WITH SECURITY APPROACHES	
	2.2 RESEARCH GAPS	
	2.3 PROPOSED SYSTEM	
	2.3.1 OVERVIEW OF THE PROPOSED SYSTEM	
	2.3.2 OBJECTIVES OF THE PROPOSED SYSTEM	
	2.3.3 ADVANTAGES OF PROPOSED SYSTEM	
3	**SYSTEM DESIGN**	14
	3.1 SYSTEM ARCHITECTURE	
	3.2 MODULE DESIGN	
	3.3 DATA FLOW / UML DIAGRAMS	
	3.4 THREAT MODEL AND ATTACK SURFACE	
	3.5 SECURITY CONTROLS	
	3.6 LAYERED VIEW OF THE ARCHITECTURE	
	3.7 ALERT DATA MODEL	
	3.8 TRACEABILITY OF OBJECTIVES TO DESIGN	

4	**SYSTEM DESCRIPTION**	26
	4.1 DATA COLLECTION	
	4.1.1 INSTALLATION LAYOUT OF THE SMART RFID EMERGENCY POLE	
	4.1.2 ACQUISITION AND TIMING PARAMETERS	
	4.2 DATA PREPROCESSING	
	4.2.1 ALGORITHMS / METHODOLOGY	
	4.2.2 RESULTS AND DISCUSSION	
	4.2.3 GRAPHICAL ANALYSIS OF RESULTS	
	4.2.4 TEST SCENARIOS AND OBSERVED OUTCOMES	
	4.2.5 DEPLOYMENT COST PROJECTION	
	4.2.6 LIMITATIONS AND MITIGATION	
5	**CONCLUSION**	41
	5.1 KEY FINDINGS AND CONTRIBUTIONS	
	5.2 ROADMAP FOR FUTURE ENHANCEMENTS	
	REFERENCES	44
	APPENDIX	46

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
1.1	SDG ALIGNMENT	4
1.2	CONVENTIONAL EMERGENCY REPORTING VERSUS SINGAPEN AI	6
2.1	COMPARISON WITH REPRESENTATIVE EXISTING SYSTEMS	9
2.2	COMPARISON OF THE THREE ALERT CHANNELS USED IN SINGAPEN AI	10
3.1	CORE HARDWARE MODULES – SMART RFID EMERGENCY POLE	16
3.2	THREAT ANALYSIS AND MITIGATION	22
3.3	SECURITY CONTROLS BY COMMUNICATION CHANNEL	23
3.4	LOGICAL FIELDS OF THE ALERT RECORD	25
3.5	TRACEABILITY OF PROJECT OBJECTIVES TO DESIGN FEATURES	25
4.1	KEY PERFORMANCE METRICS	27
4.2	ACQUISITION AND OPERATING PARAMETERS	30
4.3	TEST SCENARIOS AND OBSERVED OUTCOMES	38
4.4	ESTIMATED COST PER KILOMETRE OF CORRIDOR	38
4.5	OBSERVED LIMITATIONS AND MITIGATION	39
5.1	FUTURE ENHANCEMENTS, BENEFITS AND CHALLENGES	41

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO.
1.1	CONVENTIONAL EMERGENCY REPORTING VERSUS THE SINGAPEN AI RESPONSE FLOW	5
2.1	POSITIONING OF SINGAPEN AI IN THE REVIEWED RESEARCH LANDSCAPE	12
3.1	SINGAPEN AI SYSTEM ARCHITECTURE	14
3.2	HARDWARE BLOCK DIAGRAM OF THE SINGAPEN AI SYSTEM	15
3.3	SEQUENCE DIAGRAM FOR RFID TRIGGER	17
3.4	USE CASE DIAGRAM OF THE SINGAPEN AI SYSTEM	18
3.5	STATE DIAGRAM OF THE ESP32 FIRMWARE	19
3.6	FLOWCHART OF ALERT DETECTION AND MULTI-CHANNEL TRANSMISSION	20
3.7	LEVEL-1 DATA FLOW DIAGRAM OF THE SYSTEM	21
3.8	ATTACK SURFACE OF THE ALERT PATH WITH THREAT MARKERS T1–T7	22
3.9	LAYERED VIEW OF THE SINGAPEN AI ARCHITECTURE	24
4.1	PORTABLE GUARDIAN SOS DEVICE	27
4.2	COMPONENT ARRANGEMENT ON THE SMART RFID EMERGENCY POLE	28
4.3	FRONT VIEW OF SINGAPEN AI PROTOTYPE DEVICE	29
4.4	BACKSIDE VIEW OF SINGAPEN AI PROTOTYPE	29
4.5	EDGE VISION PIPELINE FOR WEAPON-CLASS VERIFICATION	32
4.6	SCHEMATIC LAYOUT OF THE DISPATCHER DASHBOARD	33
4.7	AI-BASED WEAPON DETECTION ALERT TRIGGERED AUTOMATICALLY WITH LOCATION	34
4.8	SOS ALERT TRIGGER AND DASHBOARD INDICATOR WITH LOCATION	34
4.9	END-TO-END ALERT LATENCY BY CHANNEL	37
4.10	DELIVERY SUCCESS OF THE LORA AND GSM CHANNELS	37
5.1	SUGGESTED PHASING OF FUTURE ENHANCEMENTS	42

SDG ALIGNMENT

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 5 – Gender Equality	5.2 – Eliminate violence against women and girls in public and private spaces	Portable Guardian SOS device and RFID Emergency Pole provide a low-friction panic trigger with sub-second alert delivery to route incidents to the nearest response unit, including all women police stations.
2	SDG 11 – Sustainable Cities and Communities	11.7 – Provide universal access to safe, inclusive and accessible public spaces	Embedding identity, AI threat verification, and multi-channel alerting into existing streetlight infrastructure improves the safety of public spaces at a per-unit cost realistic for municipal and campus budgets.

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

In urban areas, safety incidents frequently go unnoticed, and this delays emergency response. This especially affects women, students, and elderly residents navigating poorly lit or isolated stretches of public space. Today, a victim or witness must locate a phone, recall an emergency contact number, place a call, and then state their exact location an entire sequence of manual steps that itself accounts for the response delay. When no bystander is present, or the victim is physically unable to act, the process breaks down completely.

Passive infrastructure such as conventional CCTV networks arguably compounds the problem rather than solving it, since footage is recorded and archived for later review, offering little value during the incident itself. SingaPen AI was conceived specifically to close this intervention gap by folding emergency identification and alerting directly into the lampposts already present on nearly every street corner. Each pole-mounted unit integrates an RFID identity scanner, a manual SOS button, GPS tracking, and a camera performing AI-based threat verification. Once triggered, the system dispatches a verified alert to a command center within roughly one second, carrying GPS coordinates and, where applicable, an AI-flagged threat image.

Three field observations shaped the design of the system. First, dependence on a single wireless channel is risky, since both cellular networks and WiFi tend to degrade under crowd density, and losing that one channel means losing the alert entirely. Second, a passive camera with no human trigger cannot register subjective fear a person may sense danger well before any weapon becomes visible, and an object-detection model has no concept of that fear. Third, cost dictates deployability: in the Indian context, per-unit price is arguably the largest obstacle to scaling this kind of hardware, so the system had to be assembled from commodity, off-the-shelf components costing only a few thousand rupees per unit.

1.2 PROBLEM STATEMENT

Most existing public safety devices optimize for a single dimension of the problem either communication reach, as with GSM/LoRa trackers, passive monitoring, as with CCTV networks lacking any panic mechanism, or access control alone, as with standalone RFID systems. Rarely do these approaches combine all three with fault-tolerant transmission, and almost none are priced for large-scale municipal rollout.

A genuinely deployable system must satisfy four requirements simultaneously: it needs a low-friction physical trigger that lets someone raise an alarm instantly; where a camera is present, it must automatically verify the threat using AI rather than simply recording it; it must transmit the alert across multiple independent networks so that a single network error does not stop the broadcast; and it must remain affordable enough that covering every 50 to 100 metres along a transit corridor is financially realistic.

1.3 OBJECTIVES

- Design and build a functional lamp-post-integrated emergency pole combining RFID identification, a manual SOS trigger, GPS tracking, and YOLO-based AI verification.
- Develop a companion portable SOS device that shares the identical alert schema so that it can accompany an individual anywhere.
- Implement a three-channel transmission layer (WiFi/Firebase, LoRa, GSM) that degrades gracefully rather than failing outright when a network drops.
- Build a Flask-based command center dashboard capable of mapping incidents in real time.
- Empirically evaluate the hardware for latency, reliability, and AI detection accuracy under realistic operating conditions.

1.4 SCOPE

The scope of this project covers the design, prototyping, and campus-level evaluation of both hardware forms of SingaPen AI the fixed Smart RFID Emergency Pole and the Portable Guardian SOS device along with the supporting cloud backend (Flask REST API, Firebase Realtime Database) and the Leaflet-based dispatcher dashboard. The work includes the RFID identity pathway, manual SOS trigger, GPS-based localization, the on-device YOLO11n weapon-detection pipeline, and the redundant WiFi/LoRa/GSM

transmission layer, together with empirical measurement of latency, packet delivery ratio, detection accuracy, and battery life. The project is limited to a single-hop LoRa link and a prototype bill of materials validated through a campus deployment at Rajalakshmi Institute of Technology, Chennai; large-scale municipal rollout, mesh/relay LoRa topologies for extended range, and regulatory certification of the hardware are identified as future scope rather than part of the current deliverable.

1.5 SECURITY REQUIREMENTS / SECURITY CONSIDERATIONS

Several specific threats were considered during design. Alert spoofing is mitigated through per-device authentication tokens verified by the Flask API. RFID cards are inherently easy to clone, so the card is treated only as a hint of identity rather than a full authentication factor, with UIDs cross-checked server-side. WiFi routers were assumed likely to fail or face jamming, motivating the LoRa and GSM fallback paths. To guard against GPS interception over the air, the Firebase link is TLS encrypted, and LoRa payloads are kept minimal enough that little of value could be extracted even if intercepted.

Traffic crossing the WiFi network is encrypted under TLS 1.2 or later via the Firebase SDK. Stored alerts and profiles in the cloud are protected by Firebase security rules, restricting access to authenticated dispatchers only. Because LoRa and GSM operate under tight bandwidth constraints, the full payload is not encrypted on those channels; instead, only a bare device ID and coordinate pair is transmitted, with the server resolving complete incident details internally once the initial signal arrives. Because the system tracks location and captures visual data from public spaces, deployment requires clear signage indicating an AI-monitored area, strict data-retention limits to avoid indefinitely storing footage of ordinary daily activity, dashboard access restricted to authorized dispatchers only, and informed consent for any RFID-linked profile data, limited strictly to the minimum information required to assist someone during an emergency.

1.6 SDG ALIGNMENT

The project aligns with United Nations Sustainable Development Goal 5 (Gender Equality) and Goal 11 (Sustainable Cities and Communities), and its subscription-free, open design was chosen deliberately to keep the system accessible regardless of socio-economic status.

TABLE 1.1 SDG ALIGNMENT

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 5 – Gender Equality	5.2 – Eliminate violence against women and girls in public and private spaces	Portable Guardian SOS device and RFID Emergency Pole provide a low-friction panic trigger with sub-second alert delivery to route incidents to the nearest response unit, including all-women police stations.
2	SDG 11 – Sustainable Cities and Communities	11.7 – Provide universal access to safe, inclusive and accessible public spaces	Embedding identity, AI threat verification, and multi-channel alerting into existing streetlight infrastructure improves the safety of public spaces at a per-unit cost realistic for municipal and campus budgets.

1.7 COMPARISON OF CONVENTIONAL AND PROPOSED RESPONSE

To appreciate the intervention gap that SingaPen AI addresses, it is useful to compare the sequence of actions a person must complete in a conventional emergency report with the sequence required by the proposed system. In the conventional flow, shown in Fig. 1.1(a), the victim must first recognize the threat, retrieve and unlock a phone, dial an emergency number, wait for the call to be answered, and describe both the incident and a

location that may not be known precisely. Each step depends on the victim remaining calm and physically able to act, and each one adds delay before a responder is even informed.

In the proposed flow, shown in Fig. 1.1(b), the same person performs a single deliberate action: tapping an RFID card on a pole or pressing the SOS button on the portable device. Location capture, payload construction, multi-channel transmission, and map display are then carried out automatically, and the camera on the pole can independently raise an AI-corroborated alert without any human input at all.

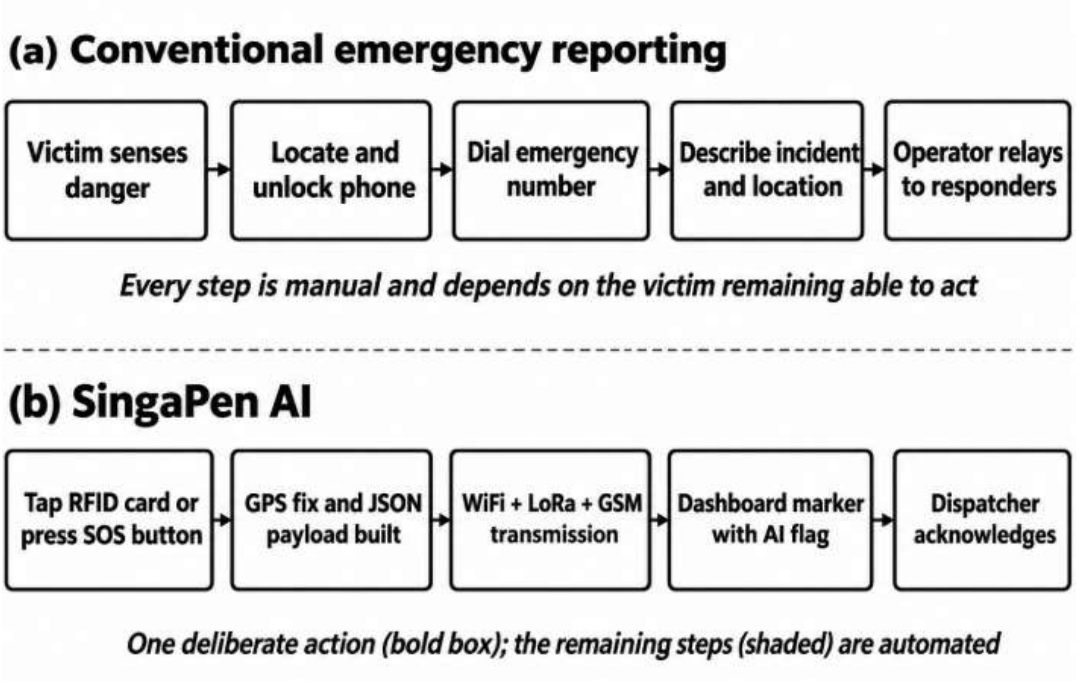


Fig. 1.1 Conventional Emergency Reporting versus the SingaPen AI Response Flow

Table 1.2 contrasts the two approaches across the dimensions that matter most to a dispatcher: the effort demanded of the victim, how location and identity are established, whether the threat is verified, how many networks the alert can use, and how long it takes to reach the control room. Values quoted for SingaPen AI are those measured in the campus evaluation reported in Chapter 4.

TABLE 1.2 CONVENTIONAL EMERGENCY REPORTING VERSUS SINGAPEN AI

Aspect	Conventional call	SingaPen AI
Trigger effort	Unlock phone, dial a number, wait for the call to connect	A single RFID tap or SOS press (3-second long press on the portable unit)
Location	Described verbally by the caller	GPS coordinates attached automatically (2.8 m CEP)
Identity	Depends on the caller	RFID-linked profile retrieved from Firebase, with consent
Threat verification	None until a responder arrives	YOLO11n weapon-class flag (confidence ≥ 0.20), reviewed by the dispatcher
Network dependence	A single cellular call	Three independent channels: WiFi/Firebase, LoRa, GSM
Time to reach control room	Depends on the caller and the call queue	≈ 803 ms (WiFi), ≈ 1,253 ms (LoRa), 6–8 s (GSM SMS)
Requirement on victim	Must be able to speak and act	Minimal action; the camera can raise an alert without any button press

1.8 ORGANIZATION OF THE REPORT

The remainder of this report is organized into four further chapters, a list of references, and an appendix.

- Chapter 2 (Literature Review) surveys wearable and pole-mounted safety devices, AI-based weapon detection, LoRa emergency communication, and smart-pole research, identifies the gaps in existing work, and introduces the proposed system.
- Chapter 3 (System Design) presents the system architecture, module design, data flow and UML diagrams, threat model, security controls, layered view, alert data model, and the traceability of design features to project objectives.
- Chapter 4 (System Description) describes data collection on the pole and the portable device, the vision pipeline and mathematical formulation, and the experimental results, together with test scenarios, a cost projection, and limitations.
- Chapter 5 (Conclusion) summarizes the key findings and sets out a roadmap of future enhancements.
- The Appendix covers applications, cost and maintenance, ethical considerations, a sample alert payload and a commissioning checklist.

CHAPTER 2

LITERATURE REVIEW

2.1 EXISTING SYSTEM WITH SECURITY APPROACHES

2.1.1 Wearable and Pole-Mounted Personal Safety Devices

Wearable and pole-mounted personal safety systems have been studied extensively. A common design pattern couples a GPS receiver with a GSM module so that a device can transmit its location as a text message [1]. Many implementations rely on an ESP32 or Arduino board together with a NEO-6M GPS unit and a SIM800-series GSM module, sending coordinates to a predefined contact list upon a button press. Some researchers have pursued hands-free triggering through biometric inputs such as heart-rate monitors [1], where a sudden spike is used as the SOS trigger. Other prototypes combine a physical button with GSM, GPS, and WiFi to notify both personal contacts and local police [2]. A related approach pairs an ESP32 camera with motion and acoustic sensors to capture images automatically during a distress event [3]. Several survey papers catalogue these GPS- and GSM-based combinations in depth [4]. Another design direction employs combined chipsets such as the SIM808, handling GSM and GPS together and often pairing them with accelerometers to detect physical altercations without requiring the user to press anything [5]. A recurring limitation across this body of work, however, is reliance on a single communication path, leaving these devices vulnerable to a single point of failure.

2.1.2 AI-Based Weapon Detection Systems

A second body of relevant work concerns automated weapon detection through neural networks. The YOLO (You Only Look Once) architecture reframed object detection as a single-pass inference task, making real-time video analysis practical for the first time [6]. Successive versions from YOLOv3 through YOLOv8 have since been deployed across IoT surveillance networks [7]. Some teams have successfully run YOLOv4 pipelines on constrained edge hardware [8], while others trained YOLOv5 specifically on custom knife and firearm datasets [9]. Comparative studies of YOLOv7 and YOLOv8 report measurable gains in mean average precision for weapon classes [10]. Additional work has focused on compressing YOLO variants for severely resource-limited hardware [11]. A separate line

of inquiry moves away from visible-spectrum cameras entirely, exploring thermal and millimeter-wave imaging for concealed weapon detection [12], [13], which remains largely orthogonal to the visible-spectrum camera used in this project but is a promising area for future extension.

2.1.3 LoRa-Based Emergency Communication

LoRa is widely favored for emergency communication owing to its low power draw and independence from cellular infrastructure. Some projects have built self-powered, hand-cranked LoRa devices with integrated GPS for disaster-response scenarios [14]. Others have assembled LoRa mesh networks from inexpensive ESP32 and SX1262/SX1278 hardware, demonstrating self-healing behavior without requiring a central LoRaWAN gateway. Trigger-based configurations have also been explored, where a microcontroller transmits a LoRa signal directly to a receiver station that raises an alert on receipt, proving that reliable single-hop delivery is achievable without any internet dependency [15]. Solar-powered LoRa mesh deployments extend this further, sustaining operation entirely off-grid. Collectively, this literature supports the choice of LoRa as a dependable backup channel alongside GSM and WiFi.

2.1.4 Smart Pole and Street-Light Integration

A final relevant thread concerns smart-pole research, which demonstrates that embedding electronics and sensors into street-lighting infrastructure is practically viable. Existing systems perform adaptive lighting control and fault detection, while others incorporate a panic button paired with a cloud-connected camera. Some designs go further, converting the pole into a broader utility hub equipped with solar panels and environmental sensors. This project extends that line of work by introducing RFID-based identity tracking together with on-device AI threat verification, a combination that, to the best of available knowledge, had not previously appeared in street-lighting platforms.

TABLE 2.1 COMPARISON WITH REPRESENTATIVE EXISTING SYSTEMS

System Class	Comm. Channels	AI Verif.	RFID ID	Approx. Cost
GPS/GSM wearable [2]	GSM only	No	No	Low
Biometric wearable [1]	GSM	No	No	Moderate
CCTV + panic pole	WiFi/cloud	No	No	Moderate
YOLO surveillance [9]	Network	Yes	No	Mod-High
LoRa disaster node [15]	LoRa only	No	No	Low
Smart utility pole	WiFi/cloud	No	No	High
SingaPen AI (proposed)	WiFi + LoRa + GSM	Yes	Yes	Low

2.1.5 Communication Technologies for Emergency Alerting

The choice of radio technology largely determines whether an emergency alert arrives at all. WiFi offers high throughput and low latency but is limited to the coverage of an access point and depends on the router and its backhaul remaining operational. Cellular GSM offers wide coverage through existing infrastructure and works with a bare SIM card, but SMS delivery is store-and-forward and its latency depends on carrier load, as the 6–8 s delivery times in this work confirm. LoRa, a chirp-spread-spectrum modulation used in the sub-GHz ISM bands, trades data rate for range and receiver sensitivity, which makes it well suited to short alert frames such as a device identifier and a coordinate pair. Table 2.2 summarizes the properties of the three channels used by SingaPen AI, with measured values from Chapter 4 where available.

TABLE 2.2 COMPARISON OF THE THREE ALERT CHANNELS USED IN SINGAPEN AI

Channel	Role in the system	Coverage	Data rate	Infrastructure dependence	Measured latency
WiFi / Firebase	Primary route; carries the full payload including profile and AI flag	Access-point range (tens of metres)	High (Mbps class)	Router and internet backhaul	≈ 803 ms
LoRa 868 MHz	Cellular-independent backup using a minimal frame	Kilometres in line of sight (100% at 1 km, 85% at 5 km measured)	Low (kbps class)	Own receiver only; no carrier network	≈ 1,253 ms
GSM SMS	Last-resort notification to saved emergency contacts	Cellular coverage area	Very low; 160 characters per SMS	Carrier network	6–8 s (7,303 ms average)

Because the three channels fail for different reasons (router outage or jamming for WiFi, range and line of sight for LoRa, carrier congestion for GSM), transmitting on all of them simultaneously makes the alert path considerably more robust than any single-channel design in the surveyed literature.

2.1.6 Privacy, Ethics and Governance of Public-Space Sensing

Placing cameras and identity readers in public spaces raises legitimate privacy and governance questions, and a safety system should address them by design rather than as an afterthought. Widely accepted baseline expectations include data minimization, purpose limitation, informed consent for any personal profile, defined retention limits for imagery, and role-based access for operators. In India, the Digital Personal Data Protection Act, 2023 formalizes obligations around consent and purpose for personal data, which makes such design choices a compliance matter as well as an ethical one. Most of the reviewed prototypes concentrate on technical feasibility and say little about these safeguards. SingaPen AI therefore treats signage, retention limits, minimal RFID profiles, and dispatcher-only dashboard access as explicit design requirements (Section 1.5 and Appendix A.3).

2.2 RESEARCH GAPS

- Existing wearable and pole-mounted safety devices predominantly rely on a single communication path (GSM-only or LoRa-only), leaving them vulnerable to a single point of failure when that network degrades or fails under crowd density.
- AI-based weapon-detection literature is largely deployed as passive network-connected surveillance, without a parallel human-triggerable pathway that can register a threat before it becomes visually apparent to a camera.
- Smart-pole and street-light integration research demonstrates lighting control, fault detection, and basic panic buttons, but does not combine RFID-based identity linkage with on-device AI verification and redundant multi-channel alerting in a single unit.
- Very little of the reviewed literature reports approximate per-unit cost, making it difficult to judge whether these designs are realistic for large-scale municipal or campus deployment; SingaPen AI addresses this by keeping the bill of materials within a few thousand rupees per unit while still offering three independent channels and AI corroboration.

Fig. 2.1 places the reviewed work in three overlapping areas: multi-channel alert communication, AI visual verification, and identity or human-triggered alerting. The reviewed systems typically occupy one of these areas, or at most two, whereas SingaPen AI is designed to combine all three in a single low-cost unit.

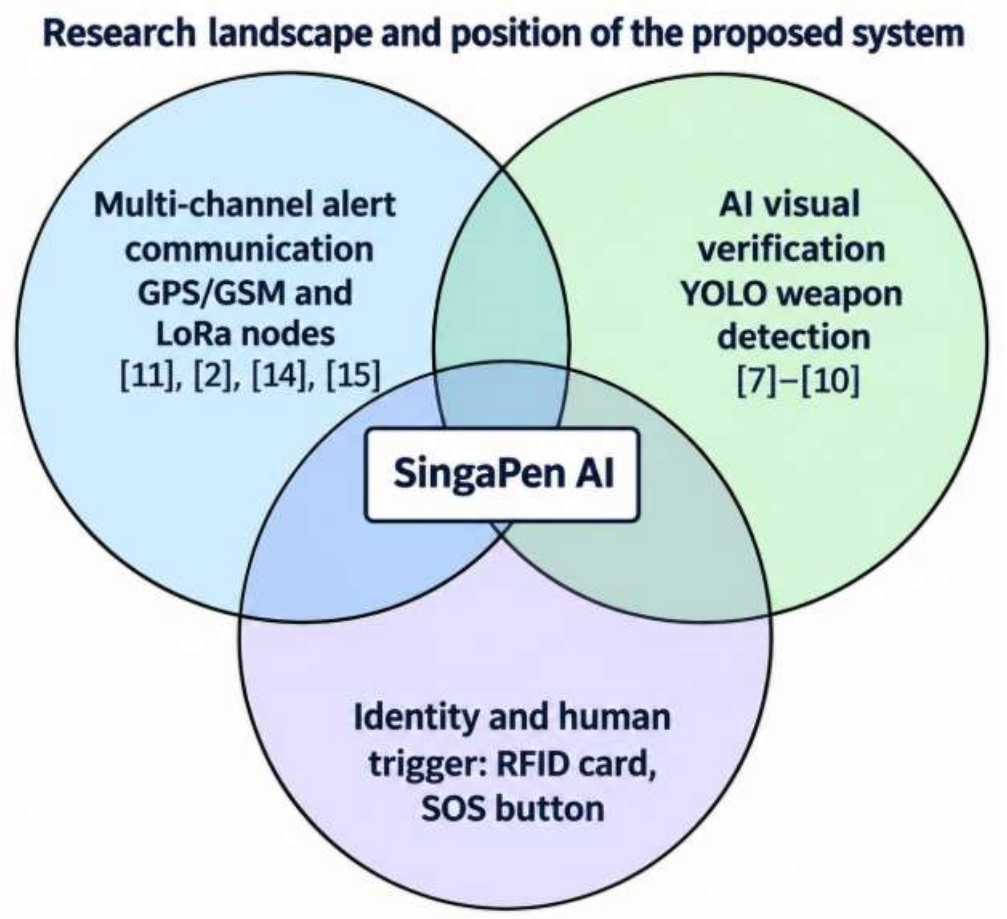


Fig. 2.1 Positioning of SingaPen AI in the Reviewed Research Landscape

2.3 PROPOSED SYSTEM

2.3.1 Overview of the Proposed System

SingaPen AI is composed of three interconnected parts. The Smart RFID Emergency Pole forms the fixed hardware layer, housing the microcontroller, RFID reader, camera, and radio modules. The Portable Guardian SOS Device is a battery-powered wearable built around the same data schema. The cloud layer, backed by Firebase Realtime Database and a Flask application, powers the Leaflet-based command center map that ties both hardware forms together. A human-triggerable RFID or SOS pathway runs in parallel to the AI verification layer, and transfer happens across three fully independent channels — WiFi/Firebase, LoRa, and GSM — so that no single network outage can silence an alert.

2.3.2 Objectives of the Proposed System

The proposed system aims to: (1) pair RFID-based identity with AI visual confirmation so dispatchers receive a preliminary threat assessment rather than a bare map

marker; (2) provide a redundant three-channel transmission system with empirically measured delivery ratio and latency; (3) unify the alert schema and cloud backend across both hardware forms, so alerts from a fixed pole or a portable wearable converge on the same dashboard; and (4) keep the bill of materials remarkably low — the portable device costing approximately Rs. 2,125 — while preserving full multi-channel redundancy.

2.3.3 Advantages of Proposed System

- Relative to single-channel GPS/GSM wearables, SingaPen AI offers two fully independent fallback channels (LoRa and GSM) alongside an automated AI corroboration step that most comparable designs omit.
- Compared with standard YOLO-based surveillance setups, the addition of a physical, human-triggerable SOS pathway combined with RFID identity linkage represents a meaningful improvement, since a person may sense danger well before any weapon becomes visible to a camera.
- The redundant channel design degrades gracefully: WiFi/Firebase remains the fastest primary route (≈ 803 ms), LoRa provides a dependable cellular-independent alternative (≈ 1,253 ms), and GSM SMS, though slower (6–8 s), still reaches emergency contacts when both other options are inaccessible.
- The bill of materials keeps unit cost within reach of municipal and campus budgets, with no recurring subscription fees since the design uses standard GSM billing, the Firebase free tier, and open-source firmware.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The pole unit is built around an ESP32 dual-core microcontroller functioning as the central controller. An MFRC522 module handles 13.56 MHz RFID cards, allowing identity-linked emergency triggers. A NEO-6M GPS receiver delivers location information, while a SIM800L GSM module sends SMS notifications and an SX1278 LoRa transceiver operating at 868 MHz provides a long-range backup communication channel. A camera module delivers frames for AI processing, and power is controlled via a buck converter that ensures a consistent 5V/3.3V power rail. All incoming data is directed to a Flask backend; Firebase Realtime Database collects the live notification feed, while a Leaflet-based map shows real-time incidents for dispatchers.

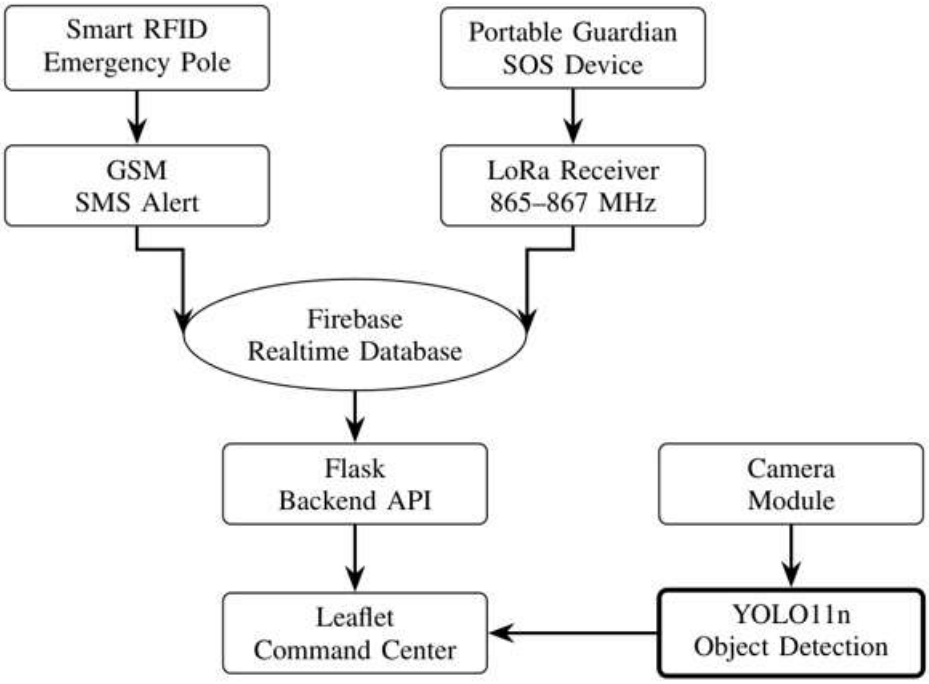


Fig. 3.1 SingaPen AI System Architecture

3.2 MODULE DESIGN

The ESP32 firmware runs a non-blocking event loop that continuously polls the RFID reader, monitors the SOS GPIO interrupt, and serializes outgoing alerts as JSON. On the server side, a Flask application provides REST endpoints to receive incoming alerts, process dispatcher acknowledgments, and serve dashboard queries. Firebase Realtime Database was selected specifically to deliver low-latency updates to linked clients. The dashboard is a standalone Leaflet.js application that connects to Firebase and displays incident markers, real-time camera thumbnails, and AI threat badges directly on the map.

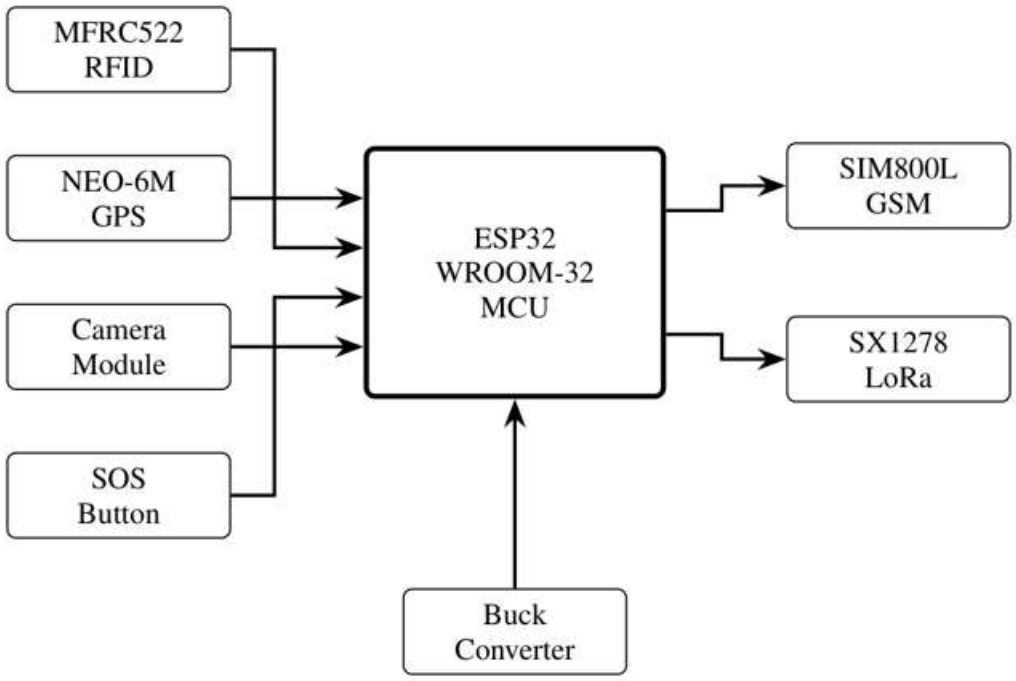


Fig. 3.2 Hardware Block Diagram of the SingaPen AI System

TABLE 3.1 CORE HARDWARE MODULES – SMART RFID EMERGENCY POLE

Module	Function	Interface
ESP32	Central controller	-
MFRC522	RFID identification	SPI
NEO-6M	GPS geolocation	UART
SIM800L	GSM SMS alerting	UART
SX1278	LoRa 868 MHz link	SPI
Camera module	Image acquisition for AI	Serial/CSI
Buck converter	Power regulation	-
Speaker/mic	Audio confirmation	I2S/analog
SOS button	Manual trigger	GPIO interrupt

Flask REST API

- POST /api/alert – allows hardware to push a new JSON payload.
- GET /api/alerts/active – lets the dashboard retrieve unresolved incidents.
- POST /api/alert/{id}/ack – handles dispatcher acknowledgment of a case.
- GET /api/device/{id}/status – provides a basic health check.

Firebase Database Schema

- devices/{device_id} – tracks registration and heartbeat status.
- alerts/{alert_id} – stores timestamps, GPS coordinates, trigger type, and the AI verification flag.
- users/{uid} – holds RFID-linked profiles so dispatchers can identify who requires assistance.

3.3 DATA FLOW / UML DIAGRAMS

The RFID trigger flow, shown in Fig. 3.3, is straightforward: a person taps their card, the ESP32 reads the UID, the corresponding profile is retrieved from Firebase, and the

resulting JSON alert is pushed to Firebase, which in turn places a marker on the dispatcher dashboard.

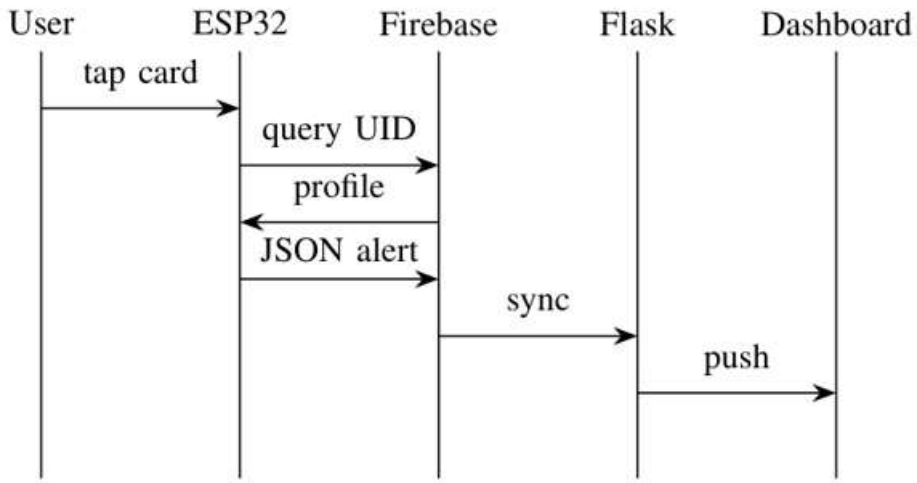


Fig. 3.3 Sequence Diagram for RFID Trigger

Three actors are defined within the system: the Civilian, the AI Verification Engine, and the Dispatcher. Civilians raise alerts through the card reader or the SOS button, while the AI engine can independently flag a threat. The dispatcher monitors the dashboard, acknowledging and resolving incidents as they arrive. The standard activity flow proceeds as:

→Trigger → Validate → Collect GPS → Construct Payload → Transmit (across three channels simultaneously) → Cloud Update → Dashboard Notify → Dispatcher Response → Incident Resolution.

The ESP32 firmware cycles through four states. It begins in Standby, polling sensors while conserving power. Upon detecting a trigger, it transitions to Triggered to handle debouncing and authentication, then moves to Transmitting to broadcast the JSON payload across all radios. It finally enters Confirmed, awaiting acknowledgment or a timeout, before returning to Standby.

3.3.1 Use Case Diagram

Fig. 3.4 shows the use cases supported by the system and the actors that take part in them. The Civilian raises alerts by tapping an RFID card or pressing the SOS button, while the AI Verification Engine independently verifies weapon classes in camera frames. Each of these triggers includes the common step of transmitting the alert over the available

channels. The Dispatcher, the only actor with access to the dashboard, views incidents on the map and acknowledges and resolves them.

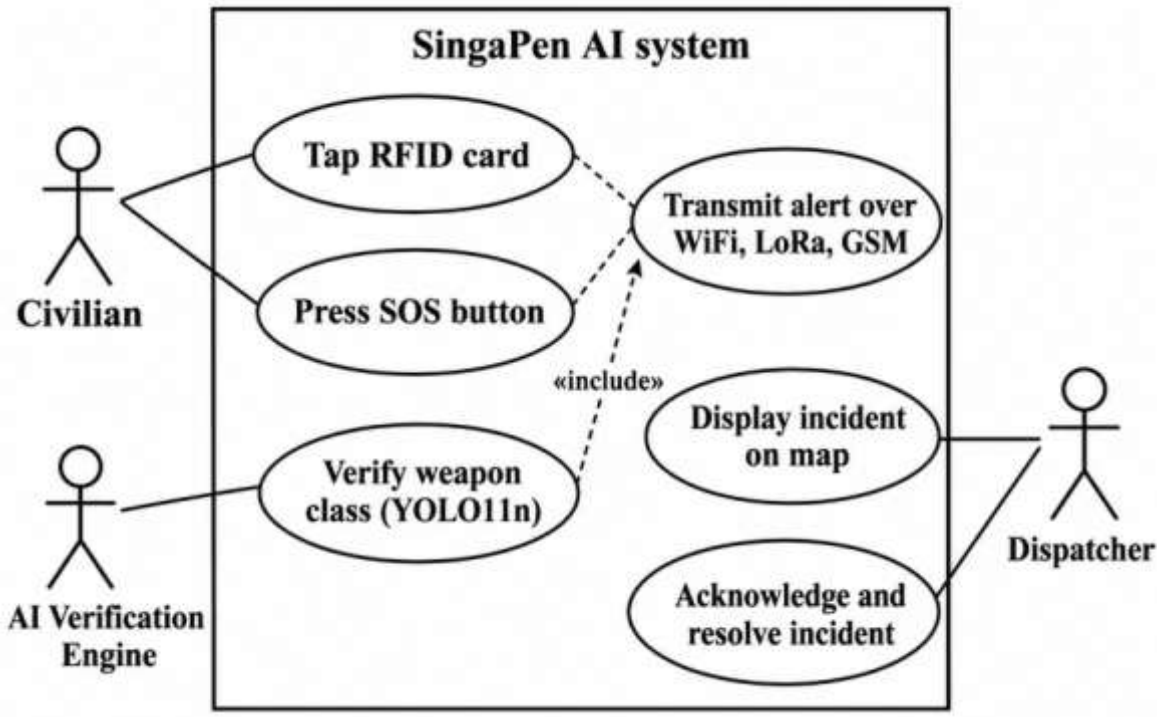


Fig. 3.4 Use Case Diagram of the SingaPen AI System

3.3.2 Firmware State Diagram

The four firmware states described above are summarized in Fig. 3.5. Standby is the default state, in which the device conserves power while polling the RFID reader and the SOS interrupt. A valid trigger moves the firmware to Triggered, where the input is debounced and authenticated; a contact bounce or an unreadable card returns it directly to Standby. In Transmitting, the JSON payload is sent on all radios, and in Confirmed the device waits for an acknowledgement or a timeout before returning to Standby. Keeping the state machine this small makes the behavior predictable and easy to verify.

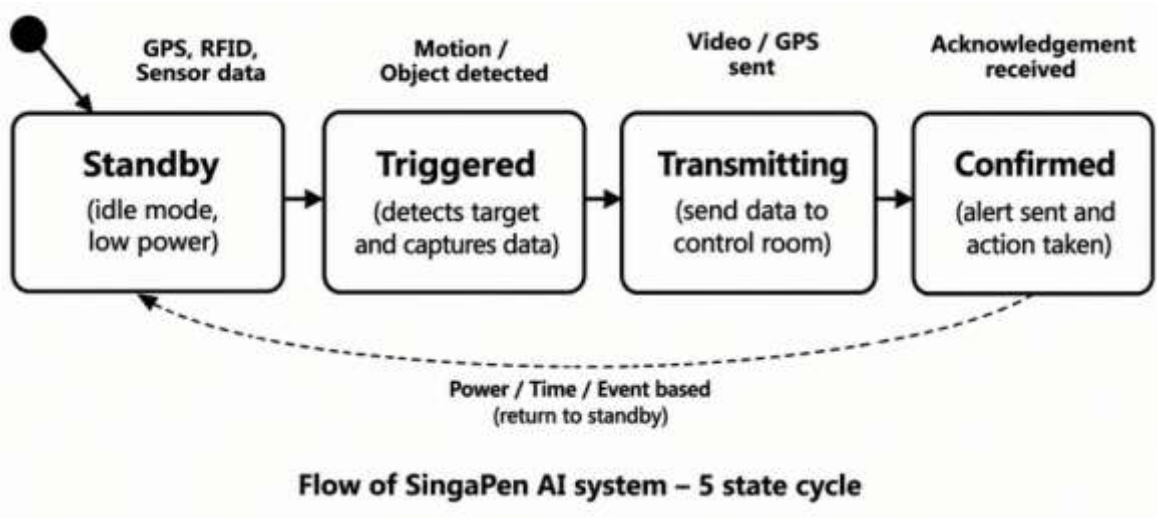


Fig. 3.5 State Diagram of the ESP32 Firmware

3.3.3 Alert Processing Flowchart

Fig. 3.6 expands the activity flow described earlier into a flowchart. After a trigger is detected and validated, the device obtains a GPS fix (using the most recent cached fix on the portable unit to save time), assembles the JSON payload, and transmits it on the WiFi/Firebase, LoRa, and GSM paths in parallel. Because the paths are independent, the loss of any one of them does not prevent the alert from reaching the control room through another. The cloud update creates the dashboard marker, and the device returns to Standby once the dispatcher acknowledges the alert or the timeout expires.

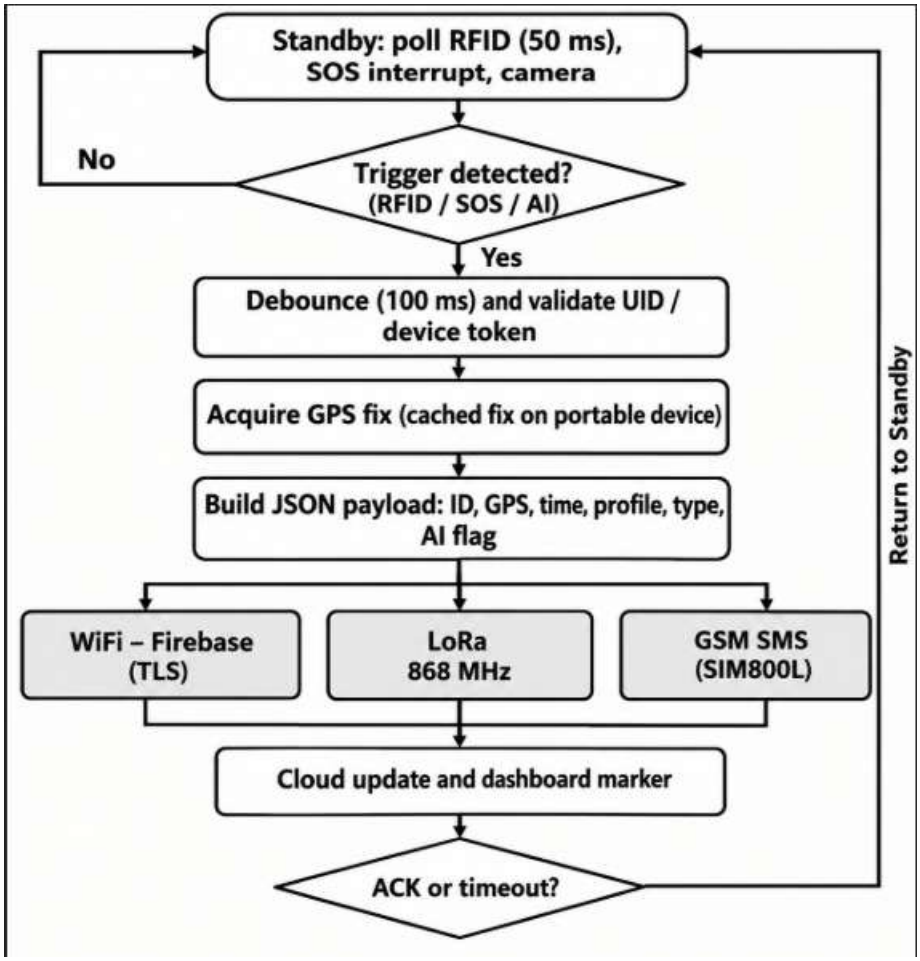


Fig. 3.6 Flowchart of Alert Detection and Multi-Channel Transmission

3.3.4 Data Flow Diagram

The movement of data between external entities, processes, and data stores is shown as a Level-1 data flow diagram in Fig. 3.7. Trigger events from the civilian (process 1.0) and AI flags from the camera pipeline (process 2.0) converge on process 3.0, which builds and transmits the alert. Process 4.0 stores alerts and serves them to the dispatcher interface (process 5.0), while the users store supports profile lookup for RFID-linked identities.

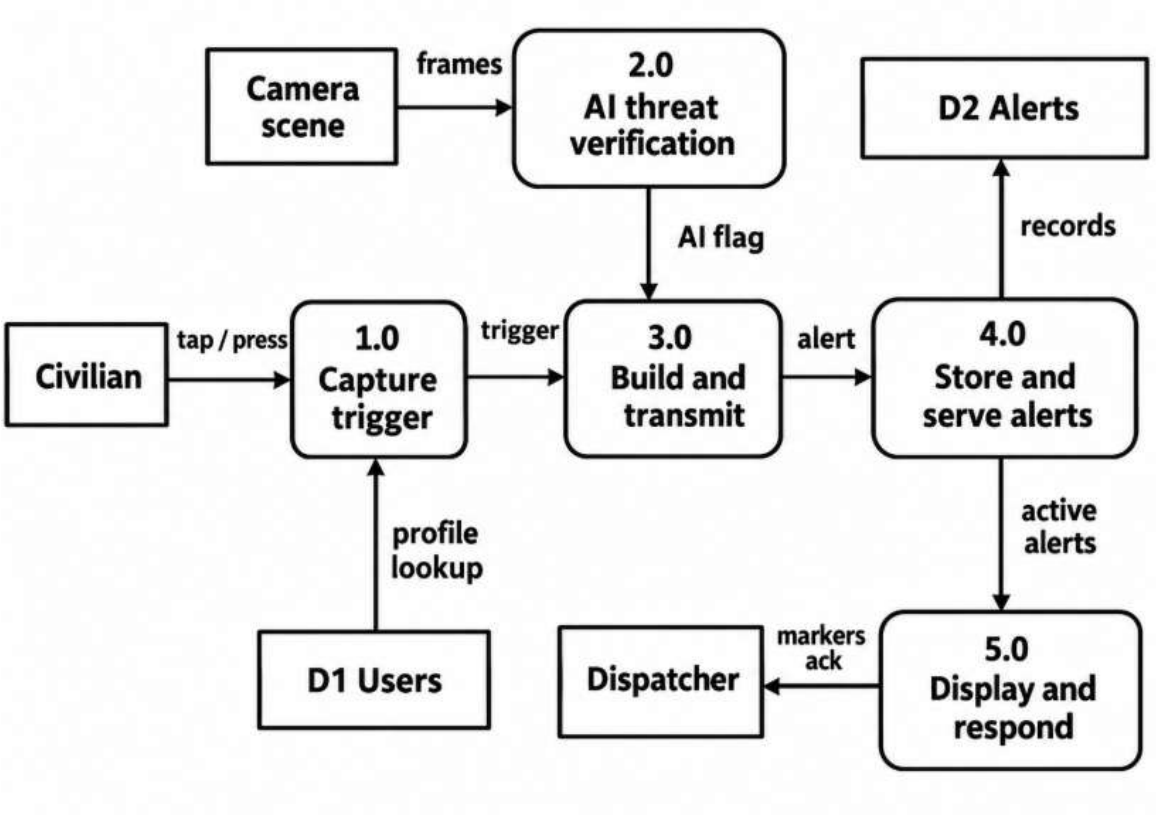


Fig. 3.7 Level-1 Data Flow Diagram of the System

3.4 THREAT MODEL AND ATTACK SURFACE

Several specific threats were considered during design. Alert spoofing is mitigated through per-device authentication tokens verified by the Flask API. RFID cards are inherently easy to clone, so the card is treated only as a hint of identity rather than a full authentication factor, with UIDs cross-checked server-side an attacker in possession of a cloned card can trigger an alert but cannot assume a verified identity. WiFi routers were assumed likely to fail or face jamming, which motivated the LoRa and GSM fallback paths as an availability control rather than purely a reliability feature. The camera and its inference pipeline represent a further attack surface: a false-positive-inducing object could be used to trigger nuisance alerts, which is mitigated by keeping a human dispatcher in the loop rather than allowing the system to auto-dispatch a response.

Table 3.3 lists the threats identified during the design review, the possible impact of each, and the mitigation adopted. The identifiers T1 to T7 are also marked on the attack-surface view in Fig. 3.8, which maps each threat to the component or link it targets.

TABLE 3.2 THREAT ANALYSIS AND MITIGATION

ID	Threat	Impact	Mitigation
T1	Alert spoofing through forged calls to the Flask API	False incidents; wasted patrol effort	Per-device authentication tokens verified by the Flask API; Firebase security rules
T2	RFID card cloning	Alert raised under another person's identity	Card treated as a hint of identity, not an authentication factor; UIDs cross-checked server-side
T3	WiFi router failure or jamming	Primary channel lost	LoRa and GSM fallback paths act as an availability control
T4	Look-alike objects used to trigger nuisance AI alerts	Alert fatigue for dispatchers	Human dispatcher reviews every alert; no automatic dispatch
T5	Eavesdropping on LoRa or GSM traffic	Exposure of device location	Minimal payload (device ID, coordinates, alert type); details resolved server-side
T6	Unauthorized access to the dashboard	Disclosure of incidents and profiles	Firebase rules restrict access to authenticated dispatchers only
T7	Over-retention of footage and profile data	Privacy intrusion	Strict retention limits, signage, and minimal consent-based profiles

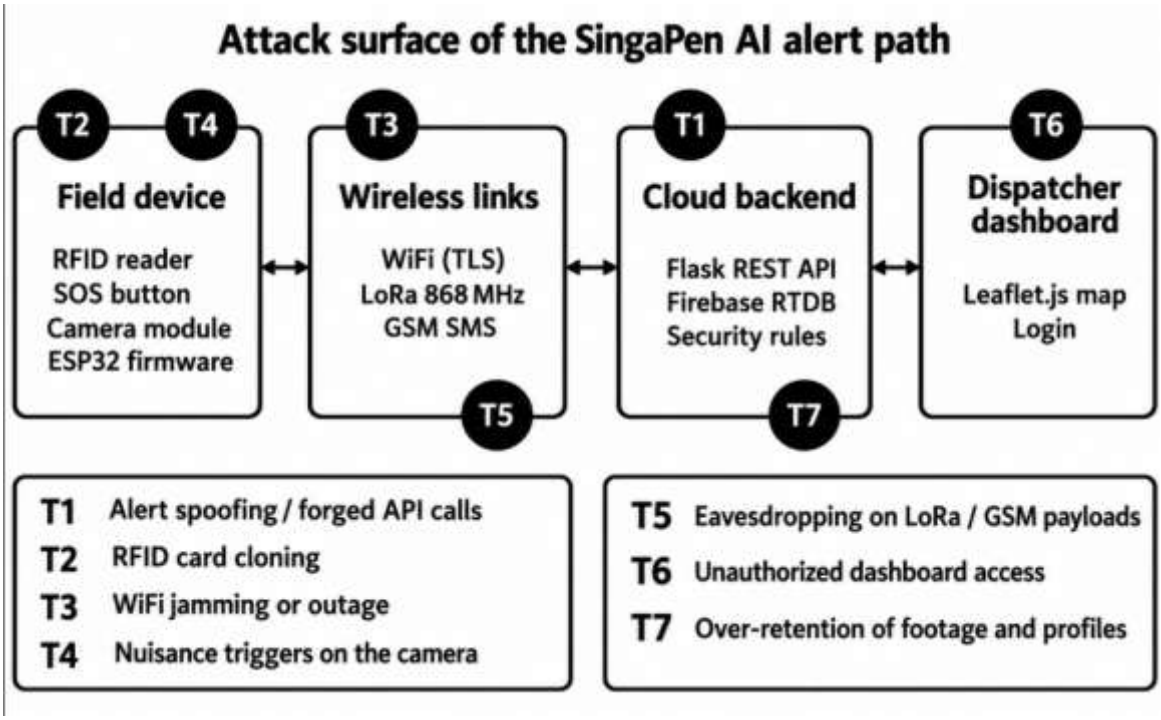


Fig. 3.8 Attack Surface of the Alert Path with Threat Markers T1–T7

3.5 SECURITY CONTROLS

The WiFi link to Firebase relies on standard TLS-secured HTTPS, offering a well-established balance of safety and robustness. The LoRa connection is managed differently: transmission takes place over the 868 MHz ISM frequency using the SX1278 module, and due to LoRa bandwidth constraints, a streamlined custom framing layer was created that sends only the device ID, GPS location, and a concise alert-type identifier. Through GSM, the ESP32 transmits standard AT commands via UART to the SIM800L module, dispatching preformatted SMS messages to saved contacts.

Traffic crossing the WiFi network is encrypted under TLS 1.2 or later via the Firebase SDK. Stored alerts and profiles in the cloud are protected by Firebase security rules, restricting access to authenticated dispatchers only. Because LoRa and GSM operate under tight bandwidth constraints, the full payload is not encrypted on those channels; instead, only a bare device ID and coordinate pair is transmitted, with the server resolving complete incident details internally once that initial signal arrives.

The controls applied to each channel differ because the channels differ in bandwidth and in the trust that can be placed in them. Table 3.3 summarizes the protection applied, the content carried, and the residual risk that remains on each path.

TABLE 3.3 SECURITY CONTROLS BY COMMUNICATION CHANNEL

Channel	Protection applied	Content carried	Residual risk
WiFi / Firebase	TLS 1.2 or later via the Firebase SDK; Firebase security rules; per-device token	Full alert record: device ID, GPS, timestamp, profile reference, alert type, AI flag	Depends on router and backhaul; susceptible to jamming
LoRa 868 MHz	Streamlined custom framing; minimal payload; full-payload encryption omitted because of bandwidth limits	Device ID, GPS location, alert-type identifier	Interception reveals little; single-hop range is limited
GSM SMS	AT commands over UART to the SIM800L; preformatted messages to saved contacts	Preformatted message with device ID and coordinates	Carrier latency of 6–8 s; SMS is not end-to-end encrypted

3.6 LAYERED VIEW OF THE ARCHITECTURE

The architecture of Section 3.1 can also be read as five layers, shown in Fig. 3.9. The perception layer contains the sensors that capture identity, manual triggers, location, and imagery. The edge-processing layer, built around the ESP32, runs the event loop, the YOLO11n inference, and the payload builder. The communication layer offers three independent paths to the cloud. The cloud layer receives alerts through the Flask API and the Firebase Realtime Database, and the application layer presents them to the dispatcher. Separating the layers in this way allows each to be improved independently; for example, the vision model can be replaced without changing the radio layer, and a new channel can be added without modifying the dashboard.

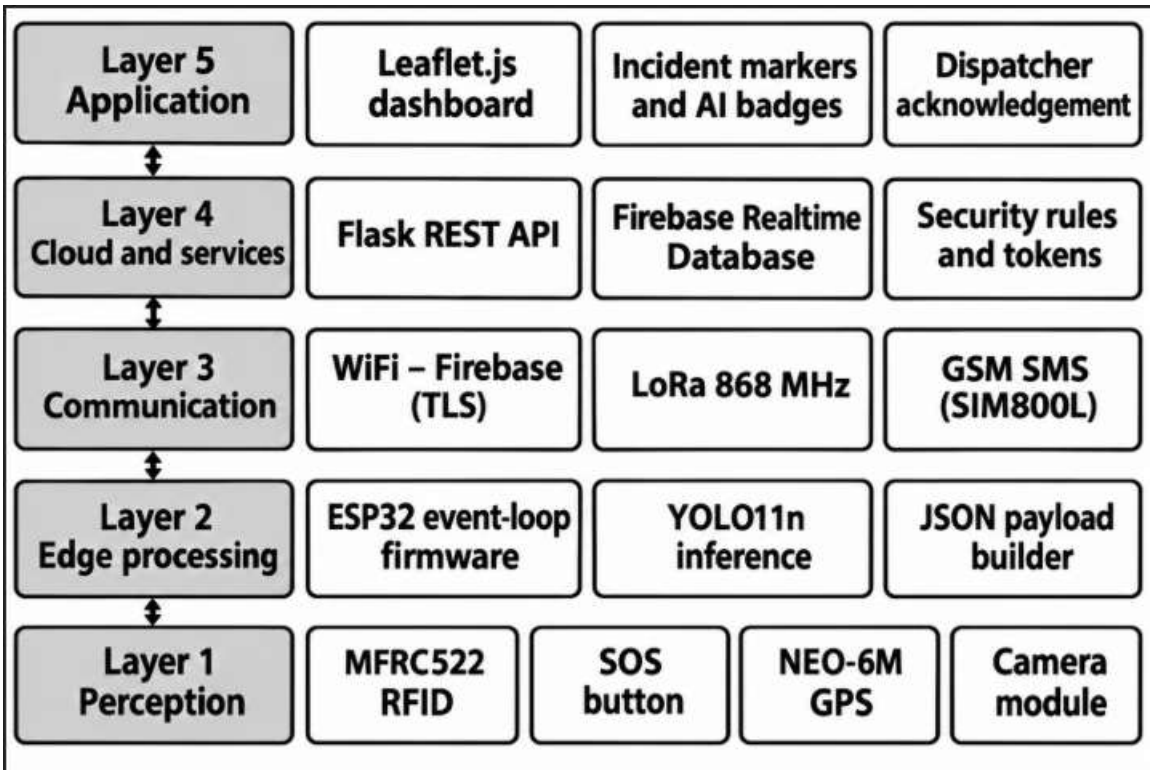


Fig. 3.9 Layered View of the SingaPen AI Architecture

3.7 ALERT DATA MODEL

Both hardware forms share one alert schema, which is what allows a fixed pole and a portable wearable to appear on the same dashboard. Table 3.4 lists the logical fields of an alert record and the channels that carry them. The full record travels over WiFi/Firebase, whereas LoRa and GSM carry only the minimal fields, and the server resolves the remaining details once the initial signal arrives.

TABLE 3.4 LOGICAL FIELDS OF THE ALERT RECORD

Field	Description	Carried over
Device ID	Identifies the pole or portable unit that raised the alert	WiFi, LoRa, GSM
GPS location	Latitude and longitude from the NEO-6M receiver	WiFi, LoRa, GSM
Alert type	Trigger source: RFID, SOS button, or AI detection	WiFi, LoRa, GSM
Timestamp	Time at which the alert was constructed	WiFi / Firebase
User profile	RFID-linked profile that helps dispatchers identify who needs assistance	WiFi / Firebase; resolved server-side for LoRa and GSM
AI verification flag	Weapon-class detection with bounding box and confidence	WiFi / Firebase
Status	Active, acknowledged, or resolved; updated by the dispatcher	Cloud only

3.8 TRACEABILITY OF OBJECTIVES TO DESIGN

To confirm that the design addresses each objective listed in Section 1.3, Table 3.5 traces every objective to the design features that realize it and to the place where the supporting evidence is presented.

TABLE 3.5 TRACEABILITY OF PROJECT OBJECTIVES TO DESIGN FEATURES

Objective	Design feature	Evidence
Lamp-post emergency pole with RFID, SOS, GPS, and YOLO verification	ESP32 controller with MFRC522, NEO-6M, camera module, and YOLO11n (Table 3.1)	Figs. 3.1 and 3.2; Section 4.2.1
Portable SOS device sharing the identical alert schema	Portable Guardian built on the ESP32 with GSM and LoRa (Table 3.2)	Section 4.1; Table 3.5
Three-channel transmission that degrades gracefully	Parallel WiFi/Firebase, LoRa, and GSM transmission	Fig. 3.6; Table 3.4
Flask-based dashboard mapping incidents in real time	Flask REST API, Firebase Realtime Database, Leaflet.js dashboard	Section 3.2; Fig. 4.6
Empirical evaluation of latency, reliability, and AI accuracy	Campus deployment with defined test scenarios	Tables 4.1 and 4.3; Figs. 4.9 and 4.10

CHAPTER 4
SYSTEM DESCRIPTION

4.1 DATA COLLECTION

Mode 1 – Smart RFID Emergency Pole: the pole spends most of its operating time in a low-power standby state, polling the RFID reader every 50 ms while monitoring the SOS button through a GPIO interrupt. When an RFID card is tapped, the RC522 reads the UID in approximately 35 ms, and the ESP32 immediately queries Firebase for the associated profile. A press of the SOS button is debounced over 100 ms to rule out accidental triggers. The NEO-6M module then acquires a GPS fix, typically under 5 seconds with location assistance, though a cold start can take up to 28 seconds. Once coordinates are available, the device ID, location, timestamp, user profile, and alert type are packaged into a JSON payload and transmitted simultaneously across LoRa (868 MHz), GSM SMS, and WiFi/Firebase. If the camera is active, YOLO11n analyzes the current frame for weapons.

Mode 2 – Portable Guardian SOS Device: the portable device follows a simple operating cycle boot, acquire GPS coordinates, and enter deep-sleep standby. A 3-second press of the SOS button wakes the device and constructs an alert, using the most recently cached GPS fix to save time before transmitting over GSM and LoRa. To assist responders in tracking the user afterward, GPS coordinates are refreshed every 30 seconds for five minutes following an alert. User feedback is kept minimal: an OLED display confirms “ALERT SENT,” and a buzzer emits three short beeps.

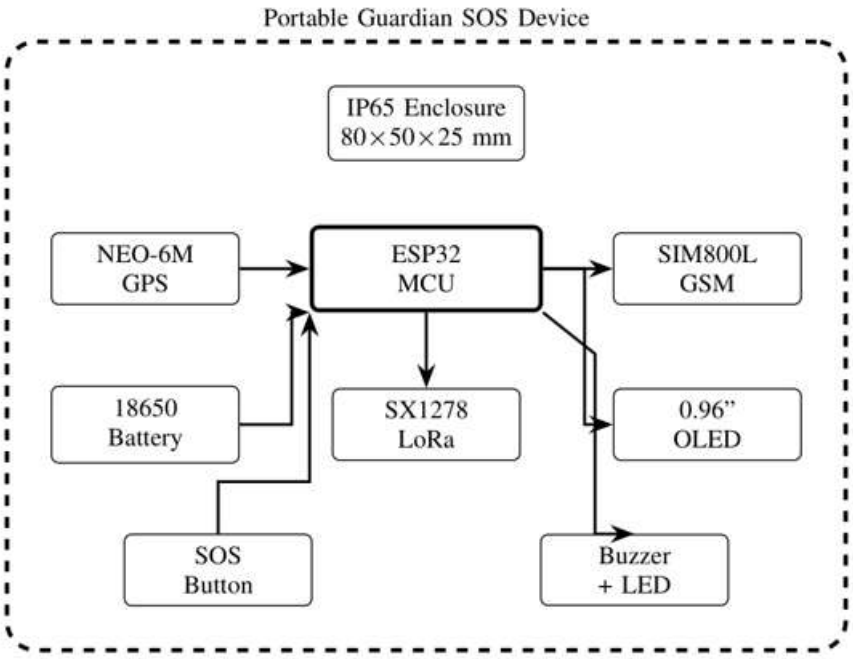


Fig. 4.1 Portable Guardian SOS Device

Three priorities guided the Portable Guardian design: simplicity of use, communication reliability, and affordability. A single SOS button was chosen deliberately to minimize cognitive load during high-stress moments, making the device usable regardless of age or technical familiarity. Redundant wireless channels (GSM, LoRa, and WiFi) ensure the alert still reaches the cloud even when one network path fails entirely. Physical dimensions settled at 80 mm × 50 mm × 25 mm and a weight of approximately 120 g. The device runs on a 3.7 V, 2600 mAh Li-Ion battery, providing roughly 48 hours of standby or 10 hours of active use, and requires about 3 hours to fully charge via the TP4056. GPS accuracy is approximately 2.5 m in autonomous mode, the SOS trigger requires a deliberate 3-second long press, and the unit is rated for operation between 10°C and +60°C within an IP65 enclosure.

4.1.1 INSTALLATION LAYOUT OF THE SMART RFID EMERGENCY POLE

Fig. 4.2 shows how the components of the fixed unit are arranged on a lamp post. In this layout the camera is mounted high on the pole for a wide field of view, whereas the RFID reader, SOS button, and speaker and microphone are placed at a height that a person can reach easily. The ESP32, NEO-6M, SIM800L, SX1278, and buck converter are housed together in a single enclosure, and the buck converter regulates the incoming supply to consistent 5 V and 3.3 V rails. The antennas for GPS, LoRa, and GSM are kept outside the enclosure so that radio performance is not degraded.

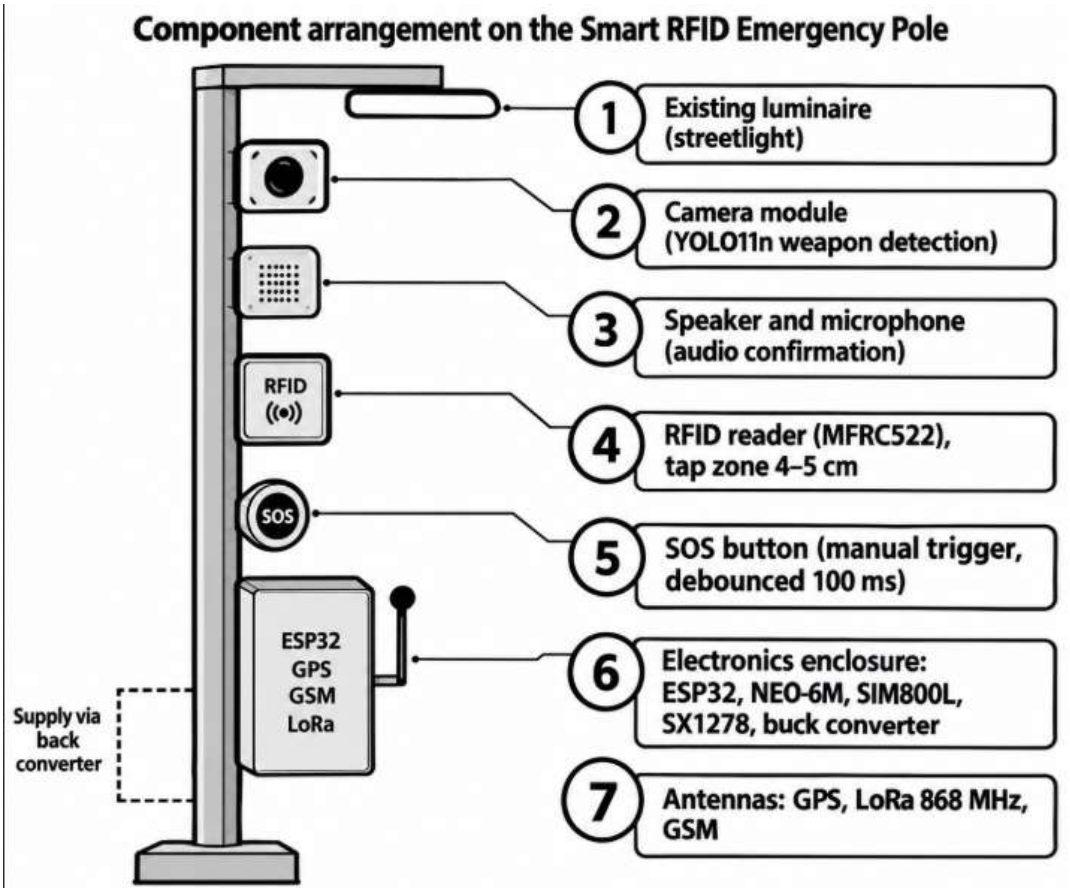


Fig. 4.2 Component Arrangement on the Smart RFID Emergency Pole (Schematic)

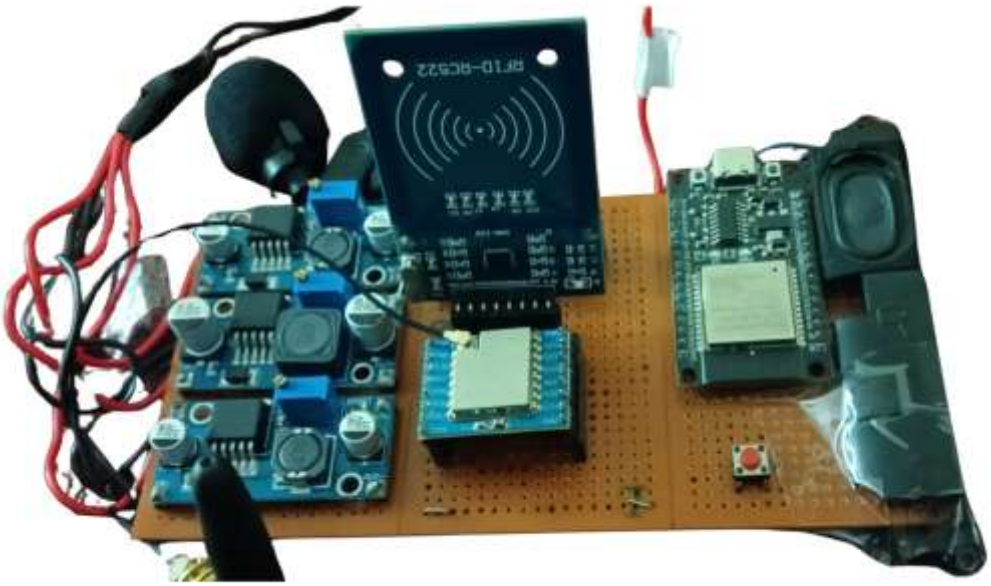


Fig. 4.3 Front view of sigapen AI prototype device

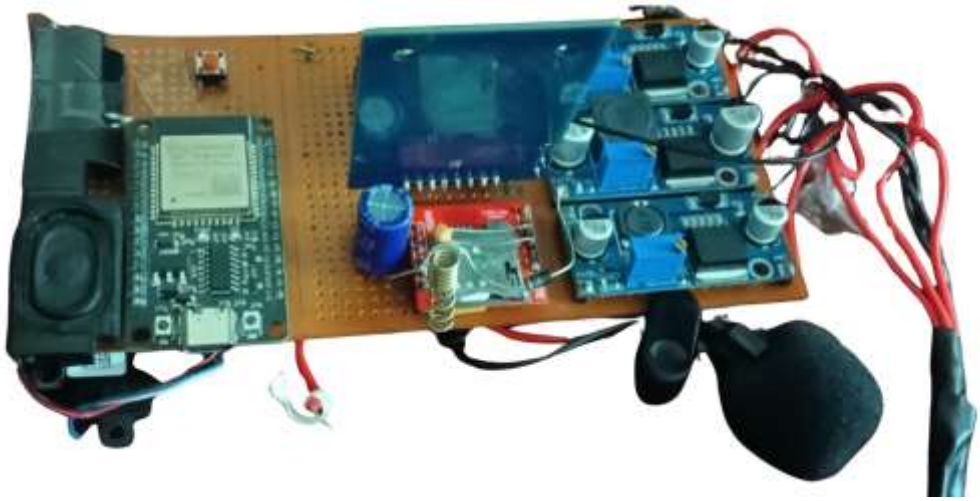


Fig 4.4 Backside view of Singapen AI prototype

4.1.2 ACQUISITION AND TIMING PARAMETERS

Table 4.2 consolidates the timing and operating parameters that govern data collection on the two hardware forms. These are the values used during the campus evaluation, and they explain the responsiveness figures reported in Section 4.2.2.

TABLE 4.2 ACQUISITION AND OPERATING PARAMETERS

Parameter	Value	Applies to
RFID polling interval	50 ms	Pole
RFID UID read time	≈ 35 ms	Pole
SOS debounce period	100 ms	Pole
SOS long-press duration	3 s	Portable device
GPS fix, assisted	< 5 s	Pole and portable device
GPS cold start	Up to 28 s	Pole and portable device
GPS refresh after an alert	Every 30 s for 5 min	Portable device
AI confidence threshold (τ)	0.20	Pole camera
Dashboard polling interval	1 s	Dashboard
Battery	3.7 V, 2600 mAh Li-Ion	Portable device
Enclosure, size, and mass	IP65, 80 × 50 × 25 mm, ≈ 120 g	Portable device

4.2 DATA PREPROCESSING

4.2.1 ALGORITHMS / METHODOLOGY

The vision pipeline is built around Ultralytics YOLO11n, a compact single-stage detector selected specifically for its ability to run efficiently on edge-adjacent hardware. Frames captured by the pole camera undergo minimal preprocessing mainly resizing and normalization — prior to being processed by the network, which outputs bounding boxes and corresponding confidence scores. Detection is limited to the COCO weapon categories that include knives (class 43) and scissors (class 76). Any detection with confidence above 0.20, the threshold set for this prototype, sets an AI-corroborated alert flag that is attached to the outgoing JSON payload. Notably, the camera can independently initiate an alert based purely on AI detection even when the SOS button is never pressed.

Pole-side alert construction (Algorithm 1): on an RFID tap, the UID is read with a 35 ms timeout and looked up in Firebase; on an SOS press, the input is debounced over 100 ms. GPS is then acquired (assisted fix < 5 s, cold start ≤ 28 s), a JSON payload is built from the device ID, GPS, timestamp, and profile, the camera frame is run through YOLO

inference if active, and the payload is transmitted over LoRa, GSM, and Firebase in sequence.

AI weapon-class verification (Algorithm 2): each frame is preprocessed and passed through YOLO11n inference; for every detection whose class is knife or scissors and whose confidence is at least the threshold $\tau = 0.20$, the detection, its bounding box, and confidence are returned as a positive AI verification.

Two mathematical models support the system. The great-circle distance between an alert and the nearest patrol unit is computed with the Haversine formula, using angular differences $\Delta\varphi$ and $\Delta\lambda$ between the alert location ($\varphi 1$, $\lambda 1$) and the responder location ($\varphi 2$, $\lambda 2$), with Earth radius $R \approx 6{,}371$ km. Detection performance is evaluated with standard metrics restricted to the weapon classes present in the test data: Precision, Recall, F1-score, Intersection over Union (IoU), and mean Average Precision (mAP). End-to-end alert latency L is defined as the cumulative delay across trigger, GPS acquisition, payload construction, transmission, cloud update, and dashboard rendering, while radio reliability is assessed via packet delivery ratio (PDR) for both LoRa and GSM, and portable-device battery life is estimated from a duty-cycled formula using active/standby current draw and capacity.

4.2.1.1 Vision Pipeline

The vision pipeline is summarized in Fig. 4.5. Each frame from the pole camera is resized and normalized to 640 × 640 pixels and passed through YOLO11n, whose outputs are filtered to the knife and scissors classes. A detection is accepted only if its confidence reaches the threshold $\tau = 0.20$. An accepted detection sets the AI-corroborated flag and attaches the bounding box to the outgoing payload, while any detection below the threshold is discarded. The deliberately low threshold favors recall, since the dispatcher reviews every flagged frame and a missed weapon is costlier than a nuisance flag.

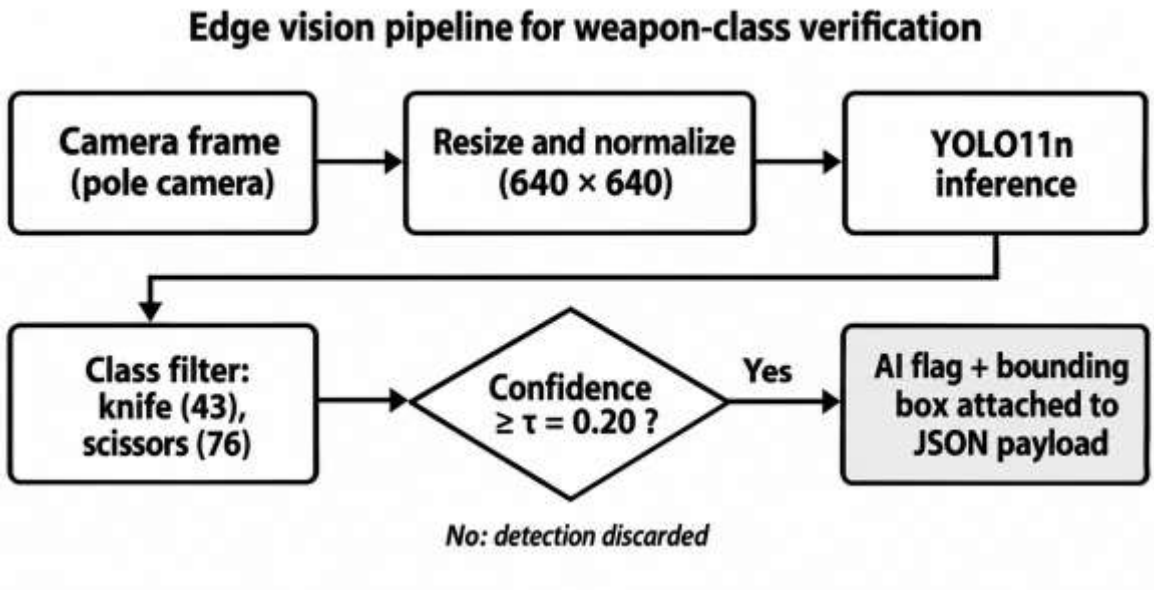


Fig. 4.5 Edge Vision Pipeline for Weapon-Class Verification

4.2.1.2 Mathematical Formulation

The quantities used to design and evaluate the system are defined below. The great-circle distance d between an alert location (φ_1, λ_1) and a responder location (φ_2, λ_2) is obtained with the Haversine formula, where $\Delta\varphi$ and $\Delta\lambda$ are the angular differences in latitude and longitude and R ≈ 6,371 km is the Earth radius:

$$a = \sin^2(\Delta\varphi / 2) + \cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2(\Delta\lambda / 2) \quad (1)$$

$$d = 2R \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)}) \quad (2)$$

Detection quality is measured with precision, recall, and F1-score, and with the intersection over union (IoU) between a predicted box B_p and a ground-truth box B_g. Mean average precision (mAP) is the mean of the per-class average precision values, restricted to the weapon classes present in the test data:

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN) \quad (3)$$

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (4)$$

$$IoU = |B_p \cap B_g| / |B_p \cup B_g| \quad (5)$$

End-to-end alert latency L is the sum of the trigger, GPS acquisition, payload construction, transmission, cloud update, and dashboard rendering delays, and the packet delivery ratio (PDR) is the percentage of transmitted packets that are received:

$$L = t_{trigger} + t_{GPS} + t_{build} + t_{tx} + t_{cloud} + t_{dash} \quad (6)$$

$$PDR = (N_{received} / N_{sent}) \times 100\ \% \quad (7)$$

For the portable device, the average current I_{avg} follows from the duty cycle D of active operation, and the battery life T_{batt} follows from the rated capacity C and a usable-capacity factor η:

$$I_{avg} = D \cdot I_{active} + (1 - D) \cdot I_{sleep} \quad (8)$$

$$T_{batt} = (C \cdot \eta) / I_{avg} \quad (9)$$

Taking C = 2,600 mAh, an average active draw of 228 mA, and $\eta \approx 0.85$ gives approximately 9.7 hours, which agrees with the active-mode battery life reported in Table 4.1.

4.2.1.3 Dispatcher Dashboard

The dispatcher dashboard, illustrated schematically in Fig. 4.6, is a single-page Leaflet.js application that reads the alert feed from Firebase at a one-second interval. Each active incident appears as a marker on the map, with distinct symbols for pole-raised and portable-device alerts, an AI badge where the vision pipeline has flagged a weapon class, and a camera thumbnail in the side panel. The dispatcher acknowledges an alert through the POST /api/alert/{id}/ack endpoint and marks it resolved once the response is complete. Because the interface presents the evidence but leaves the decision to a person, the system operates as a triage aid and never dispatches a patrol automatically.

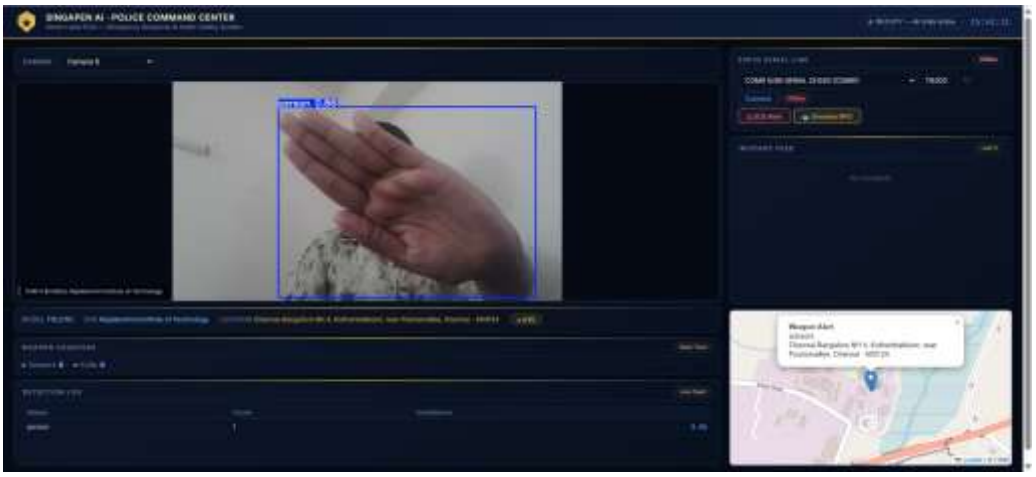


Fig. 4.6 Schematic Layout of the Dispatcher Dashboard

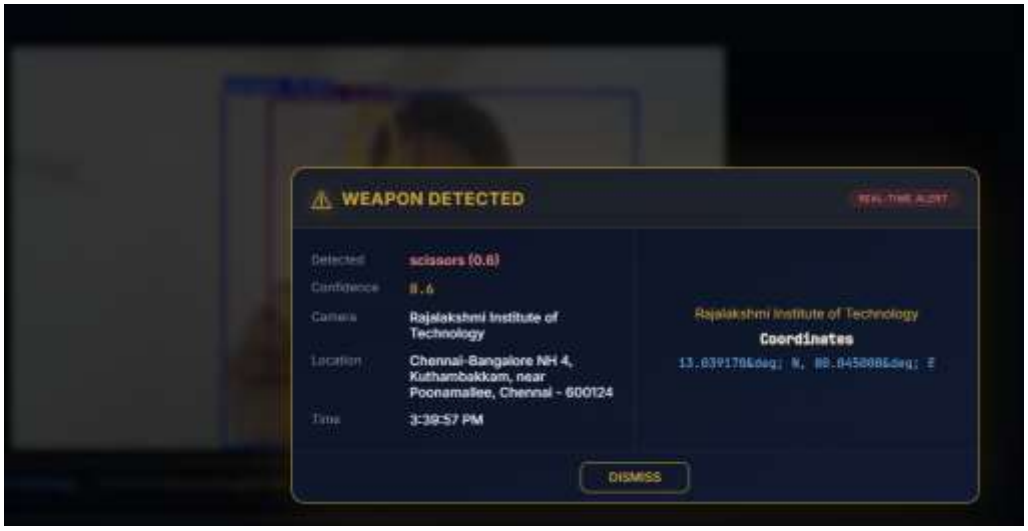


Fig 4.7 AI based weapon detection alert trigger automatically with location



Fig 4.8 SOS Alert trigger and dashboard indicator with location

4.2.2 RESULTS AND DISCUSSION

Pole firmware was written in C++ using the Arduino/ESP-IDF toolchain. RFID and LoRa peripherals were connected over SPI, while GPS and GSM modules used standard UART. The backend was implemented in Python using Flask for REST endpoints and the Firebase Admin SDK for database synchronization. The AI pipeline centers on Ultralytics YOLO11n, with OpenCV capturing and preprocessing frames. Testing was conducted on both a CPU-only development machine and a GPU-equipped workstation to capture the full range of inference latency, and a single-page Leaflet.js dashboard polled Firebase at a 1-second interval. Real-world testing took place through a campus deployment at

Rajalakshmi Engineering College, Chennai, with LoRa range tested at 1 km and 5 km line of sight, GSM delivery tested across two carriers, and AI accuracy probed with a labeled set of knife and scissors images mixed with visually similar distractor objects.

TABLE 4.1 KEY PERFORMANCE METRICS

Metric	**Value**	**Notes**
AI Inference (CPU)	45 ms/frame	YOLO11n, 640×640
AI Inference (GPU)	2.5 ms/frame	-
GPS Accuracy (CEP)	2.8 m	NEO-6M, 8–10 sats
GPS Cold Start	28 s	Assisted: < 5 s
RFID Read Range	4–5 cm	100% UID match
RFID Read Time	35 ms avg	-
LoRa 1 km LOS	100%	Packet success
LoRa 5 km LOS	85%	Packet success
GSM SMS (Carrier A)	90% delivery	8 s average
GSM SMS (Carrier B)	100% delivery	6 s average
End to End (WiFi)	803 ms	-
End to End (LoRa)	1,253 ms	-
End to End (GSM)	7,303 ms	-
Dashboard Latency	< 1 s	1 s polling
Battery (Active)	9.7 h	228 mA avg draw
Battery (Standby)	> 48 h	Deep sleep 5 µA

The WiFi/Firebase path proved to be the fastest channel by a clear margin, averaging about 803 ms, making it well suited as the primary route. LoRa required a bit more time, providing alerts at around 1,253 ms on average, yet served as a dependable, cellular-independent alternative. GSM SMS experienced significant delays, averaging between 6 to

8 seconds based on the carrier; though not ideal as a main channel, it still serves as a reliable alternative for alerting emergency contacts when other options are inaccessible.

On the AI side, the system accurately detected scissors and knives at a 0.20 confidence threshold, with a false positive rate around 5 percent, mainly caused by visually similar items like pliers and letter openers. This error rate was viewed as acceptable in practice, since a human dispatcher always examines the footage before sending out a patrol. GPS accuracy measured 2.8 m CEP, well within the precision needed for emergency dispatch, and the dashboard sustained its 1-second polling interval without degradation even under continuous video streaming.

Several limitations emerged during testing. The 6 to 8 second GSM delivery latency rules it out as a sole primary channel, so it functions strictly as a backup. LoRa packet success also dropped from a perfect 100 percent at 1 km to 85 percent at 5 km line of sight, suggesting that longer corridors would require tighter node spacing or a proper relay/mesh topology. The AI's 5 percent false-positive rate confirms that the system functions as a triage tool rather than an autonomous decision-maker, and the GPS cold-start latency of up to 28 seconds could delay a fix if the device were fully powered off, partially addressed by waking periodically for GPS updates during standby.

4.2.3 GRAPHICAL ANALYSIS OF RESULTS

The measurements in Table 4.1 are presented graphically in Fig. 4.9 and Fig. 4.10. Fig. 4.9 compares end-to-end latency on the three channels. WiFi/Firebase is the fastest at 803 ms, LoRa is about 1.6 times slower at 1,253 ms, and GSM SMS at 7,303 ms is roughly nine times slower than WiFi. The ranking matches the intended roles of the channels: WiFi as the primary route, LoRa as the cellular-independent alternative, and GSM as the last-resort notification path.

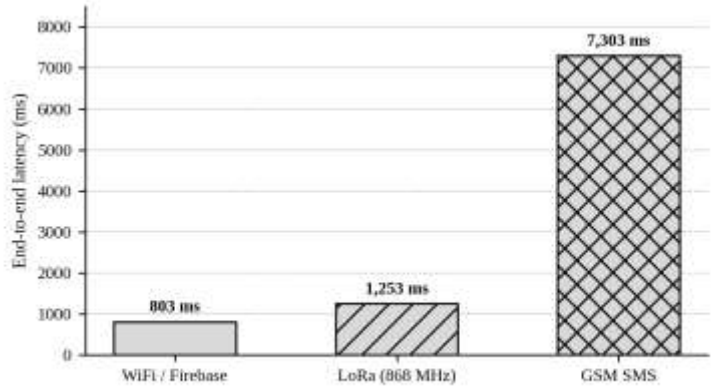


Fig. 4.9 End-to-End Alert Latency by Channel

Fig. 4.10 shows delivery success. LoRa delivered every packet at 1 km and 85 percent at 5 km in line of sight, while SMS delivery was 90 percent on Carrier A and 100 percent on Carrier B. The fall-off of LoRa with distance and the carrier-to-carrier variation in SMS delivery both argue for keeping all three channels active at once rather than relying on any single one. On the AI side, YOLO11n inference took 45 ms per 640 × 640 frame on a CPU-only machine and 2.5 ms on a GPU (Table 4.1), so even CPU-only hardware can process more than 20 frames per second.

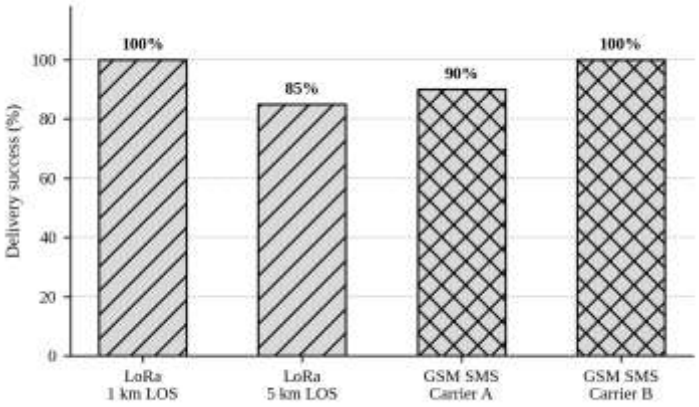


Fig. 4.10 Delivery Success of the LoRa and GSM Channels

4.2.4 TEST SCENARIOS AND OBSERVED OUTCOMES

The campus evaluation covered the ten scenarios listed in Table 4.3. For each scenario the table records the conditions under which it was run and the outcome observed, using the same measurements that appear in Table 4.1.

TABLE 4.3 TEST SCENARIOS AND OBSERVED OUTCOMES

ID	Scenario	Conditions	Observed outcome
S1	RFID identity trigger	Card tapped within read range	100% UID match at 4–5 cm; 35 ms average read time
S2	Manual SOS trigger	Button press with 100 ms debounce	Accidental contacts rejected; deliberate press raises the alert
S3	WiFi / Firebase path	Campus WiFi; dashboard polling at 1 s	803 ms average end-to-end latency
S4	LoRa link, 1 km	Line of sight	100% packet success; 1,253 ms end-to-end latency
S5	LoRa link, 5 km	Line of sight	85% packet success
S6	GSM SMS, Carrier A	Live cellular network	90% delivery; 8 s average
S7	GSM SMS, Carrier B	Live cellular network	100% delivery; 6 s average
S8	AI weapon detection	Labeled knife and scissors images mixed with look-alike objects; $\tau = 0.20$	Knives and scissors detected; false positive rate about 5% (pliers, letter openers)
S9	GPS accuracy and start-up	8–10 satellites in view	2.8 m CEP; assisted fix under 5 s; cold start 28 s
S10	Dashboard under load	Continuous video streaming; 1 s polling	Latency below 1 s with no degradation

The two weakest results, LoRa delivery at 5 km and GSM latency, are discussed as limitations in Section 4.2.6.

4.2.5 DEPLOYMENT COST PROJECTION

Because the design targets municipal and campus budgets, it is useful to project what a corridor deployment would cost. Using the prototype figures from Appendix A.2 (Rs. 9,000 to 9,500 per pole and about Rs. 500 per pole per year for maintenance), Table 4.4 estimates the cost per kilometre for the pole spacings of 50 m and 100 m mentioned in Section 1.2. The estimate covers pole hardware and maintenance only; installation labor and civil work depend on the site, and cloud hosting is assumed to remain within the Firebase free tier.

TABLE 4.4 ESTIMATED COST PER KILOMETRE OF CORRIDOR

Pole spacing	Poles per km	Hardware cost (Rs.)	Annual maintenance (Rs.)
100 m	≈ 10	90,000 – 95,000	≈ 5,000
50 m	≈ 20	180,000 – 190,000	≈ 10,000

Even the denser spacing costs less than two lakh rupees per kilometre in hardware, which is consistent with the aim of a per-unit cost that is realistic for municipal and campus budgets. For the portable device, bulk production of 100 units or more is expected to reduce the bill of materials from Rs. 2,125 to about Rs. 1,710 (Appendix A.2).

4.2.6 LIMITATIONS AND MITIGATION

The evaluation also exposed a number of limitations, which are summarized in Table 4.5 together with the mitigation adopted in the prototype or the direction proposed for future work.

TABLE 4.5 OBSERVED LIMITATIONS AND MITIGATION

Limitation	Observation	Mitigation or direction
GSM latency	6–8 s delivery, too slow for a primary channel	Used strictly as a backup to WiFi and LoRa
LoRa range	Success falls from 100% at 1 km to 85% at 5 km	Tighter node spacing or a relay/mesh topology (future work)
AI false positives	About 5%, mainly pliers and letter openers	Human dispatcher reviews every alert; used for triage, not autonomous decisions
GPS cold start	Up to 28 s when fully powered off	Periodic wake-ups for GPS refresh during standby; cached fix on the portable device
Prototype scope	Single-hop LoRa link; campus-scale validation only	Municipal rollout and regulatory certification identified as future scope

CHAPTER 5
CONCLUSION

This work demonstrates that a lamppost-integrated, RFID- and AI-verified emergency response system can be built from simple commodity hardware, at a budget realistic for schools and local municipalities, without sacrificing responsiveness. Sub-second alert latency was achieved on the primary channel alongside reliable AI weapon corroboration. The measured results 803 ms over WiFi, 1,253 ms over LoRa, and 7,303 ms via GSM, together with a 100 percent LoRa success rate at 1 km and 45 ms YOLO11n inference on CPU provide reasonable confidence that this approach could scale to real campuses and city transit systems. The portable device bill of materials totals approximately Rs. 2,125 per unit for the prototype build, and the full pole prototype setup (including camera and enclosure) costs approximately Rs. 9,000 to 9,500, with bulk production expected to bring the portable device cost down by about 20 percent.

Several extensions are planned for future work, including Whisper-based voice triggers supporting regional languages, ArcFace-based personnel verification, and integration with autonomous drone dispatch for aerial incident views. Additional directions include satellite-modem backups for fully off-grid locations, predictive crime-hotspot analytics, shifting more AI inference to the edge to reduce cloud dependency, and a companion mobile application allowing citizens to view localized alerts in real time.

5.1 KEY FINDINGS AND CONTRIBUTIONS

The main findings and contributions of this work can be summarized as follows.

- A single hardware family serves two forms of deployment: a fixed lamp-post pole and a portable wearable share one alert schema, one cloud backend, and one dashboard.
- Three independent channels give graceful degradation: alerts reached the dashboard in about 803 ms over WiFi, 1,253 ms over LoRa, and 7,303 ms over GSM, with LoRa delivering every packet at 1 km.
- On-device AI corroboration is achievable on commodity hardware: YOLO11n runs in 45 ms per frame on a CPU with a false positive rate of about 5 percent, and a human dispatcher remains in the loop.

- The design is economical: about Rs. 2,125 per portable device and Rs. 9,000 to 9,500 per pole, with no recurring subscription fees.
- Security and privacy were treated as design requirements, through token authentication, TLS, minimal payloads on constrained channels, signage, retention limits, and consent-based profiles.

5.2 ROADMAP FOR FUTURE ENHANCEMENTS

The extensions listed above differ in effort and in dependencies. Table 5.1 therefore groups them by purpose, expected benefit, and the main challenge each will face, and Fig. 5.1 suggests a phasing in which the lower-risk software and network improvements come first.

TABLE 5.1 FUTURE ENHANCEMENTS, BENEFITS AND CHALLENGES

Enhancement	Purpose	Expected benefit	Main challenge
Companion mobile application	Let citizens view localized alerts	Wider situational awareness	Authentication and alert-fatigue control
More inference at the edge	Reduce cloud dependency	Faster and more private verification	Limited compute on the pole controller
LoRa mesh and relay nodes	Extend range beyond a single hop	Better delivery beyond 1 km	Node placement and power planning
Whisper-based voice triggers	Hands-free alerting in regional languages	Works when a button cannot be reached	Noisy outdoor audio; false triggers
ArcFace personnel verification	Verify the identity of authorized personnel	Stronger identity assurance	Privacy and consent requirements
Predictive crime-hotspot analytics	Guide patrol planning	Proactive rather than reactive safety	Data quality and bias
Drone dispatch	Aerial view of an incident	Faster situational assessment	Airspace regulation and reliability
Satellite-modem backup	Connectivity at off-grid locations	Alerting where no cellular or WiFi exists	Modem cost and power draw

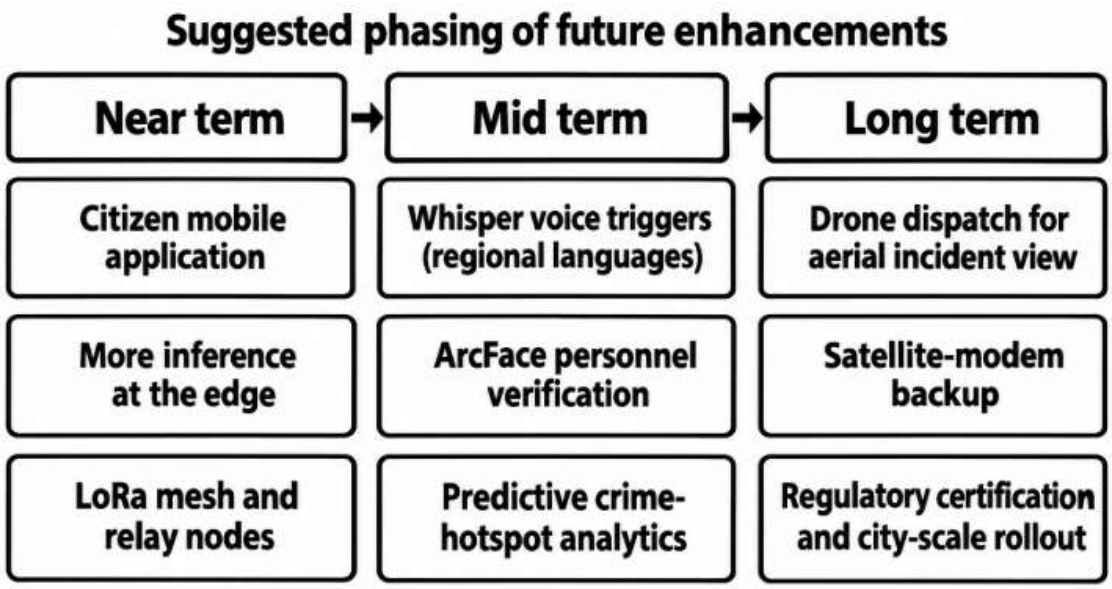


Fig. 5.1 Suggested Phasing of Future Enhancements

REFERENCES

[1] M.A. Talha, M. Hasan, M.A. Mannan, S. Ahmad, M.R. Hazari, M.S. Parvez, T. Hossain, S. Islam, IoT based women safety device using Haversine formula, in Proceedings of the 2025 4th International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), (2025) 460–464. https://doi.org/10.1109/ICREST63960.2025.10914400

[2] K. Venkatesh, S. Parthiban, P. Santhosh Kumar, C.N.S. Vinoth Kumar, IoT based unified approach for women safety alert using GSM, in Proceedings of the 2021 Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV), (2021). https://doi.org/10.1109/ICICV50876.2021.9388415

[3] S. Malaj, B. Mahesh, IoT based smart wearable device for women safety, International Journal of Engineering Technology and Management Sciences 7(6) (2023).

[4] S. Sharma, D. Salunke, E. Haldar, R. Mandlik, N. Salke, A comprehensive survey on IoT-based smart safety devices for women, in Proceedings of the 2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), (2022) 1101–1109. https://doi.org/10.1109/ICAISS55157.2022.10010784

[5] K. Sangeetha, R. Devi, A. Ananya, M.M. Suvetha, R.V. Visshnu Priya, Women safety system using IoT, in Proceedings of the 2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC), (2022) 1477–1480. https://doi.org/10.1109/ICAAIC53929.2022.9792929

[6] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: unified, real-time object detection, in Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), (2016) 779–788. https://doi.org/10.1109/CVPR.2016.91

[7] S. Narejo, B. Pandey, D. Esenarro Vargas, C. Rodriguez, M.R. Anjum, Weapon detection using YOLO V3 for smart surveillance system, Mathematical Problems in Engineering 2021(9975700) (2021) 1–9. https://doi.org/10.1155/2021/9975700

[8] A. Singh, T. Anand, S. Sharma, P. Singh, IoT based weapons detection system for surveillance and security using YOLOv4, in Proceedings of the 2021 6th International Conference on Communication and Electronics Systems (ICCES), (2021) 488–493. https://doi.org/10.1109/ICCES51350.2021.9489224

[9] A. Rehman, L.G. Fahad, Real time detection of knives and firearms using deep learning, in Proceedings of the 2022 24th International Multitopic Conference (INMIC), (2022) 1–6. https://doi.org/10.1109/INMIC56986.2022.9972915

[10] Y.Y. Aung, K.Z. Oo, Detection of guns and knives images based on YOLO v7, in Proceedings of the 2024 3rd International Conference on Artificial Intelligence for Internet of Things (AIIoT), (2024) 1–6. https://doi.org/10.1109/AIIoT58432.2024.10574710

[11] C.J. Fierro Silva, C. Del Valle Soto, C. Bran, J. Varela Aldas, Comparative analysis of previous YOLO detectors and YOLOv26s for real-time weapon detection in video surveillance, Frontiers in Computer Science 8(1789702) (2026). https://doi.org/10.3389/fcomp.2026.1789702

[12] D. Bhardwaj, A. Ramamoorthy, P. Goyal, DEF-YOLO: leveraging YOLO for concealed weapon detection in thermal imaging, arXiv preprint arXiv:2510.13326 (2025). https://doi.org/10.48550/arXiv.2510.13326

[13] L. Pang, H. Liu, Y. Chen, J. Miao, Real time concealed object detection from passive millimeter wave images based on the YOLOv3 algorithm, Sensors 20(6) (2020) 1678. https://doi.org/10.3390/s20061678

[14] R. Pradeep Kumar, K.P. Diliban Selvarathinam, C.N. Gnanaprakasam, A LoRa based emergency communication device for disaster response, in Proceedings of the 2025 5th International Conference on Trends in Material Science and Inventive Materials (ICTMIM), (2025). https://doi.org/10.1109/ICTMIM65579.2025.10988293

[15] S. Vaishnavi, R. Ragavi, V. Swathi, C. Swegha, LoRa enabled disaster management and emergency communication, International Journal of Engineering and Techniques 12(2) (2026) 1035. https://ijetjournal.org/wp-content/uploads/LoRa-Enabled-Disaster-Management-and-Emergency-Communication.pdf

[16] Espressif Systems, ESP32 Series Datasheet, Espressif Systems, Shanghai, China, 2023. https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf

APPENDIX

A.1 Applications and Impact

Several practical applications follow naturally from this system. The most direct is women's safety, whether through individually carried portable devices or fixed poles installed in isolated areas that route alerts directly to the nearest all-women police station. The system also fits well on college campuses — RFID access control and AI perimeter monitoring were piloted at Rajalakshmi Institute of Technology as part of this work. Transit hubs such as railway platforms and bus stops represent another strong use case, as do schools, where a simplified interface with basic geo-fencing and parent notification could be layered on top. Beyond these immediate cases, the architecture could be adapted for smart-city crime-hotspot analytics, hospital staff safety, senior-citizen monitoring, lone-worker safety in industry, and disaster management, especially since the LoRa channel operates as a relay even when the cellular network is completely offline.

A.2 Cost Estimation and Maintenance

The portable device bill of materials totals approximately Rs. 2,125 per unit for the prototype build. For the pole subsystem, including camera and enclosure, the full prototype setup cost approximately Rs. 9,000 to 9,500. At production volumes of 100 units or more, bulk component discounts and proper PCB integration are expected to bring the portable device cost down to roughly Rs. 1,710, a reduction of about 20 percent from prototype pricing. No recurring subscription fees were built into the design; standard GSM billing and the Firebase free tier are used throughout, and since the firmware is open source, there are no licensing costs. Estimated ongoing maintenance costs are approximately Rs. 500 per pole per year, covering battery replacement and regular visual inspections.

A.3 Ethical Considerations

Because the system tracks location and captures visual data from public spaces, ethical safeguards were a central design concern. Any real-world deployment requires clear signage indicating an AI-monitored area. Strict data-retention limits are necessary to avoid indefinitely storing footage of ordinary daily activity, and dashboard access must be restricted to authorized dispatchers only. RFID-linked profile data must also be collected

only with informed consent, limited strictly to the minimum information required to assist someone during an emergency.

A.4 Sample Alert Payload

The listing below shows an illustrative alert record as it might be pushed to POST /api/alert. Field names and values are examples only, and the coordinates are placeholders.

```
{
  "device_id": "POLE-001",
  "alert_type": "RFID",
  "timestamp": "2026-11-05T21:14:32+05:30",
  "location": { "lat": 13.0000, "lon": 80.0000 },
  "user_uid": "u_1024",
  "ai_verification": {
    "flag": true,
    "class": "knife",
    "confidence": 0.41
  },
  "status": "active"
}
```

A.5 Commissioning Checklist

Before a pole or portable unit is placed in service, the following checks are recommended.

- **Hardware:** confirm that the RFID reader returns the correct UID, that the SOS button triggers after debouncing, and that the GPS module obtains a fix.
- **Connectivity:** confirm delivery on each channel separately (WiFi/Firebase, LoRa, and GSM) and check that the dashboard shows the marker within its one-second refresh.
- **AI:** verify that the camera view is unobstructed and that the confidence threshold is set to the intended value.
- **Security:** confirm that the device token is provisioned, that TLS is in use on the WiFi link, and that Firebase security rules restrict access to authenticated dispatchers.
- **Privacy:** confirm that AI-monitoring signage is displayed and that data-retention limits are configured.
- **Operations:** confirm that dispatchers are trained to acknowledge and resolve alerts and to treat AI flags as advisory.